The Payments Perspective: Micro-foundations of Resource Analysis

Steven A. Lippman[1] and Richard P. Rumelt[1]

[1]Anderson School of Management, UCLA

Abstract

Problems in the micro-foundations of neoclassical theory, especially in partial equilibrium analysis, are being imported into the resource-based view. This paper critiques neoclassical theory and proffers new concepts more suited to strategy scholars. In particular, we develop the concepts of simple rents, rent sensitivity analysis, and the payments perspective.

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The micro-foundations of a subject are the definitions of its basic elements and the allowable operations that can be performed using these elements. Our purpose in this paper is to critique neoclassical micro-foundations that have been imported into strategy research and to proffer new concepts that, we argue, should be the micro-foundations of strategy's resource-based view.

A central element of the resource-based program is the presumptive connection between economic rent and sustainable advantage. We hold that the identification of competitive advantage with neoclassical economic rent, or with economic profit, and the associated concept of opportunity cost, imports a micro-foundations shortcoming into the strategy field. The resource-based program offers an important new way to look at the firm, but it cannot be pushed through to completion without making a clear break with the neoclassical interpretations of economic profit, economic rent, and opportunity cost.

The problem with these neoclassical economic concepts is fundamental. The problem is not

whether or not markets are perfect, or whether or not there is uncertainty or asymmetric information. The problem is at the heart of price theory itself. It is the indeterminacy in the concept of 'cost' induced by the existence of economic rents.

The purpose of this paper is to provide a critique of relevant neoclassical theory, and to sketch out a substitute framework for use in strategy research. The view we espouse is called the 'payments perspective.' Consistent with the spirit of the resource-based program, it identifies a firm's revenue

as payments for the services of its inputs and resources: Payments $\equiv$ Revenue, by definition. This point of view is called 'full imputation' and follows the ideas of Winter (1987), Friedman (1976), Buchanan (1969), and Coase (1938). For example, the revenues of a consulting firm consist of payments for commodity inputs (e.g., paper, administrative support) plus payments for the services of its specialized know-how. Crucially, there is no residual part of revenue that is not attributable to a resource. The payments to resources are what we later define as simple rents -they are neither Ricardian, transfer, differential, quasi, Paretian, compound quasi, nor economic rents. The payments perspective requires a way of identifying resource scarcity other than the concept of economic rent. To that end, we introduce rent sensitivity analysis, which permits the identification of scarce economic resources by a marginal test rather than direct reference to a wedge between value in use and value in next-best use. In a companion paper (Lippman and Rumelt, 2003) we utilize this foundation to explore the application of cooperative game theory to the questions of resource advantage and gains to trade in resources.

The paper begins with the concept of rent. The first section provides a brief history of rent concepts, and the second section uses the example of farming and land to develop and illustrate the key ideas and phenomena that economic scarcity induces. Our views on the significance of these analyses for strategy are included throughout the presentation. We end our investigation of rents with a critique of the concept of the 'supply curve' and partial equilibrium analysis. We show that there is many a slip between the 'cup' of cost and the 'lip' of price.

The paper then turns to a critique of economic cost concepts. We show that the concept of economic cost has no agreed upon definition within economics. Even Nobel Laureates disagree about these micro-foundations. We analyze and illustrate the different masters the cost concept tries to serve. In the end, we recommend a change in perspective. The language of cost obscures the simple fact that value is lodged in scarce factors, not 'economic profit.' The payments perspective helps avoid a variety of logical fallacies in reasoning about performance and value.

It is fitting in a paper about payments for value received to remark what we would owe to others, were their insights not freely given. Our thinking about the resource-based view has been shaped by the contributions of almost every scholar working in this still-small field. Winter's (1987) arguments on the history and importance of full imputation have been an important guide. Teece's (1986) work shaped our thinking about appropriation. Wernerfelt's (1984, 1995) contributions to the resourcebased view, in addition to being seminal, have been consistent sources of insight and models of clarity. Barney's (1986, 2001) contributions have pushed the frontier and uncovered important new conceptual challenges. Dierickx and Cool's (1989) paper made key contributions to ideas about the accumulation of unpriced resources. Peteraf's (1993) work on ex ante competition was an important stimulus. Mahoney's (2001) investigation of sustainable rents provided a number of valuable ideas.

Rent Concepts

The resource-based view has been the primary avenue for the transference of the concept of rent from economics to the study of competitive strategy. Bowman (1974: 47) noted that strategy was a 'search for rent' and Rumelt (1984, 1987) argued that sustained performance was due to rents, and, moreover, that the key source of rents was the coupling of uncertainty (or entrepreneurship) with isolating mechanisms. Montgomery and Wernerfelt (1988), Peteraf (1993), Mahoney and Pandian (1992), Winter (1995), and others have contributed to the identification of rents as the surplus earned by scarce, proprietary, and valuable resources.

Rent is a payment for the services of a factor of production (one pays rent for an apartment or car). An economic rent, on the other hand, is defined [1] as a payment for a factor in excess of that minimally necessary to call forth its services. Economic rent is due to scarcity: when a limited quantity of a factor is available, an increased payment (price) does not call forth an increased supply. Many authors do not bother to make the distinction between rent and economic rent, instead depending on the context to make the difference clear. For example, in the context of hiring workers, Milgrom and Roberts (1992) say, 'A rent is the portion of earnings in excess of the minimum amount needed to attract a worker to accept a particular job or a firm to enter a particular industry.'

A businessperson pays rent for land or space. This payment is a cost of doing business. But what makes such payments interesting is that they arise only because the land or space is scarce. Whereas a person or a cow or an orchard must consume resources just to exist, space exists without being fed. The payment for space is for its scarcity value. The payment rises to the level necessary to exclude others, and hence is the value of the space to others. [2]

Classical Rent

The analysis of rent was probably begun by Adam Smith (1776: 163): 'Rent . . . enters into the composition of the price of commodities in a different way from wages and profit. High or low wages and profit, are the causes of high or low price; high or low rent is the effect of it.' Forty-five years later, in a series of public arguments, Ricardo added more detail to the concept, especially clarifying the role of scarcity in rents. The point of his argument was that (Ricardo, 1821: Ch. 2, para. 19) 'If the high price of corn were the effect, and not the cause of rent, price would be proportionally influenced as rents were high or low, and rent would be a component part of price.' ... Rent does not and cannot enter in the least degree as a component part of its price.' To Ricardo, land rent was a payment determined by the land's fertility or location relative to the least favorable

[1] See 'Economic rent' in The New Palgrave (1987), entry by A. Alchian. [2] The 'exclusion' argument loses strength if there are externalities associated with resource use or when resources have the characteristics of public goods (e.g., broadcasting, software standards). When property rights are poorly defined or absent, firms depend upon a variety of isolating mechanisms to capture surplus. See Rumelt (1984).

or 'marginal' land in use. He wrote (Ricardo, 1821: Ch. 2, para. 4) : 'It is only, then, because land is not unlimited in quantity and uniform in quality, and because, in the progress of population, land of an inferior quality, or less advantageously situated, is called into cultivation, that rent is ever paid for the use of it.'

Mill (1871), like Ricardo, saw price as set by the marginal producer, who 'pays no rent.' Rents, he argued, were payments for the use of superior resources. Generalizing beyond land, Mill argued that the laws of rent applied to 'any difference in favor of certain producers' (1871: 477). He also began the exploration of the differential rent concept (below) by noting that 'when land capable of yielding rent in agriculture is applied to some other purpose, the rent which it would have yielded is an element in the cost of production of the commodity which it is employed to produce.' With this insight, Mill inaugurated the line of reasoning that sees rent as an excess payment over what a resource's payment would have otherwise been. Of course, that alternative payment is itself a payment in excess of some other payment, and so on.

Ricardo's and Mill's classical rent concept was based on two alternative measures of rent which gave the same answer: rent was measured with respect to the external as well as the internal margin. Measuring rents with respect to the external margin meant comparing the rent on a piece of land with the rent earned by zero-rent (marginal) land. Comparison on the internal margin meant comparing the marginal product of the final dose of capital and labor to the average product.

Ricardo's rent concept has a surface clarity, but glosses over subtle issues. Foremost is that he reasoned by looking at one productive use of a resource-typically corn production on land. But the alternative to growing corn is not lying fallow, but producing a different product on the same land. Consequently, 'marginal' corn land may still be paid a rent because it can be used for something else. Second, standard classical treatments imagine land of 'inferior' quality as producing less corn and hence yielding less rent. This is not necessarily true; it all depends on how the corn production function changes with 'fertility.' [3]

Pareto (1897) developed a concept of rent which has been improperly described in much modern literature. What have been called 'Paretian Rents' by Rumelt (1987) and others [4] are actually economic or transfer rents, discussed below. Pareto saw economic activity as a struggle between desires and obstacles to be overcome. He stated that under free competition two inequalities would hold: Revenue $\geq$ Cost and Price $\geq$ Marginal Cost. [5] He argued that rent exists when either inequality is strict. He said that these rents arise because of 'obstacles' to transforming savings into particular forms of capital. [6] In our language, the obstacles are barriers to imitability and substitution.

Neoclassical Rent

The present conception of 'economic rent' was formed by joining classical ideas to marginal analysis. Joan Robinson (1933: 104-107) provides a clear view of mid-neoclassical thinking about rent. She

defines transfer earnings as the payment or price 'necessary to retain a given unit of a factor in a certain industry.' The word 'necessary' signals that the payment is not the payment actually received, but what would be received in its next-best employment (plus any switching cost). She also notes that 'A unit which would remain in the industry for a smaller payment than it actually receives may be called an intra-marginal unit.' In this vein, she remarks that

> Each individual unit of each factor will be fitted into the place where its earnings will be greatest; when its earnings in that use fall it will retreat to its next most profitable use, and if there is an appreciable difference between its actual earnings and its earnings in the next most profitable use to which it could be put, it will be receiving rent.

This 'transfer rent' is also called differential [7] rent or (incorrectly) Paretian rent. Present-day economists simply call it 'economic rent.' The new concept was widely adopted by neoclassical economists because it fit neatly into the opportunity cost logic that was also being developed at the time. [8] As noted above, the standard concept in use today sees economic rent as an excess pay-ment-the portion of a payment in excess of the minimum required to transfer the resource from one use to another.

Sunflowers and Rent

Our analysis of economic rent is undertaken in a standard setting: farmers growing a homogeneous crop on a limited amount of land. We begin with the simplest possible situation and gradually add complexity. As will be seen, only a modicum of complexity is required to raise interesting questions.

Many identical tenant farmers grow highvitamin sunflowers. Each farmer takes as given the market price $P = 14$ per bundle for sunflowers. There are no scale economies or disec-onomies-the 'firms' farmers create are perfectly homogeneous. Each farm produces $b = 1.5$ bundles of sunflowers per acre. The direct cost of production is $v = 6$ per bundle. This cost includes expenses related to labor, seed, and other purchased commodities. It excludes any payments for land and any financial expenses. We shall simply call it wage cost to remind us that it adds together the money paid

[3] See Samuelson (1959) for a complete discussion. [4] The error appears to originate with Worchester (1946), who attributed to Joan Robinson (1933) the reference to Pareto. However, Robinson did not cite Pareto, attributing her term 'transfer earnings' to Henderson and to Shove (fn. 1, p. 104). Apparently following Worchester, the term 'Paretian rent' was taken up in the writings of Wessel (1967), Shepherd (1970), and others. Rumelt's (1987) use of this terminology has passed into a number of papers dealing with strategy. [5] 'Le prix de revient.' [6] Pareto (1897: Vol. II, paragraphs 745-765).

[7] Alchian (1987) uses this term, but its origin is not clear. Marshall (1920: 422) uses the term 'differential rent' to mean something much closer to differences in Ricardian rents. [8] Marshall (1920) invented the term 'quasi-rent' to describe the returns to real nonland assets. The qualification 'quasi' reflected his doctrine that in the long run all man-made inputs are variable (not scarce). Marshall also invented the term 'compound quasirent' to describe the payments to sunk investments in assets specialized to a particular supplier or business arrangement. In modern usage, the latter is termed an appropriable quasi-rent, whereas Marshall's original definition of quasi-rent is generally ignored.

for commodity inputs purchased in perfectly elastic markets. [9] Whether or not the total long-run average cost of a bundle should include the 'cost' of land is, as we shall see, in dispute.

The wage cost includes the cost of the farmer's labor and we assume that v does not vary with P. [10] Each farmer generates a net revenue of $14 - 6 = 8$ per bundle of sunflowers grown, hence each farmer's net revenue per acre is $8b = 8 \times 1.5 = 12$.

The net revenue earned by each farmer is the farm's cash profit exclusive of charges for land or financing. Because there are so many rent concepts in existence, we use the term simple rent to refer to this cash profit. Simple rent is the full rent payment, not an 'excess' payment. If the farmer owned the land, the simple rent is the cash flow that would be discounted to arrive at a valuation for the farm. Simple rent is an observable cash amount or payment.

The next assumption is that whereas farmers are unlimited in supply, there is only a limited amount of sunflower land. Furthermore, growing sunflowers is the best use of this land. Consequently, rent must be paid to the landowners for the right to farm the land. Under competition, this rent will rise to the level of net revenues (simple rent). Were it less, prospective farmers would bid more for the right to farm. Were it more, farmers would go out of business. Thus, each tenant farmer will pay 12 per acre in rent to a landlord, showing zero economic profit. The landlord's income of 12 is not an economic profit, but rather the return to a scarce factor. If the going interest rate is 6 percent per annum, the land will sell for 200 per acre and the rental income will represent a normal 6 percent return on its value.

This story is so straightforward that it hardly seems to merit further investigation. However, let us dig a little deeper and look at how the price of sunflowers is set. Assume that the amount of sunflowers demanded at a price P is given by the linear demand function $Q = Q_0(1 - P/\rho)$. For our example, $Q_0 = 500$, and $\rho = 20$. Furthermore, assume that the quantity of land available for growing sunflowers is limited to $L = 100$ acres. [11] Furthermore, assume that the land has no alternative economic use. (We soften this assumption later.)

In the absence of any shortage of land, unrestricted price competition and entry drives the price of sunflowers down to v, the wage cost per bundle. At that price, the demand for sunflowers is

$$Q_v \equiv Q_0(1 - v/\rho) = 500(1 - 6/20) = 350$$

However, the available land can produce no more than $bL = 150$ bundles of sunflowers. Hence, the limit on land becomes an active and binding constraint on production.

If all the land L is farmed at production rate b, the market price of sunflowers is

$$P_L = \rho\,(1 - bL/Q_0) = 14,$$

which is the number we proffered at the start of this exposition. The equilibrium market price of

[9] We do not call it 'variable' because a single farmer can, in the long run, alter the amount of land he uses. [10] In a more complex production function, the intensity of cultivation would depend upon the relative prices of sunflowers and labor. Pushing this complexity into the background allows a simpler exposition of the behavior of rent.

sunflowers is 14 , well above the wage cost $v = 6$ of production. Competition among the farmers will not drive the price down to wage cost per bundle because there is not enough sunflower land to support that much demand. Figure 1 displays the

industry equilibrium for this situation. As shown, the supply function for sunflowers is horizontal at price v until the land's limit of $bL = 150$ is reached, where the supply function becomes a vertical line. The market price is set where the demand curve crosses this vertical line.

Land is not scarce because it is limited in supply: it is scarce because the limitation in its supply is binding. That is, scarcity exists because $bL < Q_v$. For sunflowers, the critical level of land is $350/1.5 = 233.3$. Given 234 or more acres of land, the demand Q_v created by the lowest feasible price v can be fully accommodated. Were $L = 300$, there would be no economic (binding) scarcity of land even though its supply is fixed and limited.

The economic scarcity of land is revealed by the fact that the equilibrium price P_L of sunflowers is above wage cost v. The price of sunflowers must rise, pushing down the demand for sunflowers, until demand equals the amount the land can actually produce. The two statements 'Land is scarce and 'The price of sunflowers is above average (nonland) cost' mean exactly the same thing. This rise in the price of sunflowers is the source and essence of scarcity rent. Simple rent is the shadow-price of the binding constraint on land. The difference $R_b = P_L - v = 8$ is the simple rent per bundle of sunflowers. The simple rent per acre satisfies $R_L = b\,(P_L - v) = 12$. In terms of the parameters of the problem,

$$R_L \equiv b\,[P_L - v] = b\left[\rho\left(1 - \frac{bL}{Q_0}\right) - v\right].$$

The simple rent R_L paid for land is also 'economic rent' because, in this model (thus far), there is no alternative use for the land. Consequently, economic rent equals simple rent.

It is important to grasp the relationship between the wage cost v of production and the prices of sunflowers and land. In this simple model with homogeneous land, market price depends only upon the demand for sunflowers and the supply of land-it does not depend (at the margin) upon the wage cost v of growing sunflowers. As long as land is economically scarce, small changes in the variable cost of production have no effect [12] on the price of sunflowers (see Equation 2). However, a small increase in v has a direct negative effect on the price of land (see Equation 3). If v goes up by 1 , there is no change in the price of sunflowers, but the rent per bundle falls by 1 and the rent per acre falls by 1.5 . So the farmer faces two 'costs:' wage cost per bundle and simple rent per bundle. But the 'cost' of land behaves strangely: if wage costs are reduced the price of sunflowers stays the same and the 'cost' of land rises to compensate. The cost R_b of land per bundle doesn't rise because we are substituting land for labor or anything like that. It rises because it is simply the difference between the price of sunflowers and the wage cost. The odd behavior of land cost, we

[11] We omit 'thousands' and/or 'millions' to simplify the exposition.

[12] Technically, the wedge between variable cost and price occurs here because the marginal unit of land employed earns economic rent.

shall see, occurs because neoclassical partial-equilibrium analysis, and the cost concepts inherent to it, are only designed to determine one price at a time. But there are two prices to be determined in this problem: one for sunflowers and one for land.

Innovation and Appropriation

Thus far we have presented a very simple model of economic rent. Competition is perfect, the demand function is standard, productive units are homogeneous and small. But complexity is only a step away. It arrives with the consideration of innovation and heterogeneity.

Define a small plot of land as one whose maximum output has a negligible effect on any price. It should be clear that private innovations made by a farmer working a small plot of land increase that farmer's (or that plot's) profit. Because we will soon find conditions under which innovations do not increase value, it is useful to formalize the baseline condition.

Theorem 1: Given a small plot of land, a private increase in productivity b or decrease in wage cost v always leads to an increase in net revenue.

Proof: Before the innovation, the farm's net revenue is $R = b\,(P_L - v)$. A decrease in v adds nothing to output and clearly increases net revenue. An increase in productivity from b to $b^+ > b$ increases the farm's output, but, because the plot is small, has a negligible impact on the price of sunflowers and also on the general rental price of land in the industry. After the innovation, the farm's net revenue is $R^+ = b^+\,(P_L - v) > R$. Q.E.D.

It is, of course, not surprising that a private increase in productivity increases profit. Somewhat more interesting is the fact that the size of the impact of innovation depends upon the overall scarcity of land-on the level of simple rent. Most interesting of all is the fact that there are conditions under which a private increase in productivity will hurt a large farm (see Theorem 2 below).

Given an increase in net revenue specific to a farm, who appropriates the increased value? In neoclassical theory this question is handled by the consideration of market prices, called 'opportunity costs' in this context. Thus, if the innovation is specific to the land, such that any farmer working the land can realize it, the market price of the farmer is unchanged, but the 'opportunity cost' of the land has risen; the landowner appropriates the value. If the innovation is specific to the farmer, perhaps being specialized tacit knowledge that can be applied to any plot of similar size, similar logic points to the fact that the farmer appropriates the increase in net revenue.

What happens, however, if the innovation is specific to both the farmer and the land? That is, the innovation depends upon farmer A working plot X. If the innovation is specific both to a farmer and to a particular plot of land, neoclassical theory faces a problem. The 'opportunity cost' of the land is its market price and the 'opportunity cost' of the farmer is what he could earn in his next best employment, so the net revenue due to their complementary combination seems to be a positive economic profit.

The standard treatment of such pure profit is to attribute it to 'firm-specific' factors such as

technology or entrepreneurial skill. This solution to the 'problem' of profit was first suggested by Kaldor (1934). Friedman (1976: 149) calls this factor entrepreneurial capacity, noting that 'this term gives a name to our ignorance rather than dispelling it.' Note that this whole line of reasoning sidesteps the issue of who appropriates the extra value.

Our position on this issue is that the introduction of firm-specific 'resources' or 'rents' or 'profits' is a mistake. We believe that the spirit of the resource-based view is to impute payments to resources. In the above example, the extra payments garnered by the farm are due to the cospecialization, or interaction, between the farmer and the land. There is no mystery about the value of the co-specialized resource bundle; the problem for neoclassical theory is that the increment in value is not priced in a factor market. The creation of a mysterious firm-specific resource is simply the neoclassical economist's way of avoiding the issue of co-specialization.

The 'firm-specific' approach has two significant failings. First, it sidesteps the basic question of who appropriates the value by assuming that a firm owns the co-specialized resources. But co-specialization induces what Williamson (1985) calls the 'fundamental transformation.' This means competition is replaced with bilateral negotiation. For example, in the case of improved knowledge that is co-specialized to both the farmer and a particular piece of land, the farmer may not own the land. In that case, neither party necessarily appropriates the surplus. The division of surplus is the outcome of bilateral bargaining and the shares are formally indeterminate. We delve into this issue in greater detail in our companion paper (see Lippman and Rumelt, 2003).

Secondly, it fails badly when more than two resources are involved. With two co-specialized resources, there is only one way to combine them, and thus only one excess value. With N cospecialized resources, there are $2^N - N - 1$ different possible combinations of two or more resources to be considered and possibly valued. Taking the above example of co-specialization between a farmer and a piece of land, add more factors: a laborer who also knows the land and a buyer who is partial to the farm's product. Co-specialization immediately obliterates the assumption of alternative use values being determined by exogenous market prices.

The notion of 'firm-specific' resources is a device to avoid facing the pattern of co-specialization among resources. It is more appropriate for strategy researchers to see resources as possibly specific to one another and the firm as a legal shell holding a particular bundle or coalition of resources.

Returning to the issue of innovation, consider an increase in productivity b. That is, an imitable innovation has become known to all farmers. A first thought is that more productivity will drive down both the price of sunflowers and the simple rent paid for land. More productivity depresses the price of sunflowers, but it also increases the product per acre. Taking the derivative of R_L with respect to b in Equation 3 yields $dR_L/db = 2P_L - (v + \rho)$. From $dR_L/db = 0$ we learn that the rentmaximizing price P_L^* satisfies $P_L^* = (v+\rho)/2 = 13$. The industry demand curve $Q = Q_0(1-P/\rho)$ reveals that the rent maximizing output is

$$Q^* = Q_0 \left(1 - \frac{v+\rho}{2\rho}\right) = \frac{Q_v}{2} = 175.$$

Hence, the rent-maximizing productivity b^* satisfies $b^*L = Q_v/2$, or $b^* = Q_v/2L = 350/200 = 1.75$. Note that if $bL > Q_v$, the constraint of limited land does not bind and simple rents fall to zero. Increases in productivity above Q_v/L, therefore, induce no change in land rent. We summarize this result as [13]

> Theorem 2: There exists a $b^* > 0$ such that for $b < b^*$, an increase in productivity increases net revenue and, therefore, simple rent. On the other hand, for $b^* < b < Q_v/L$, an increase in productivity decreases net revenue and, therefore, simple rent.

The corresponding maximum rent is $R_L^* = (Q_0/\ 4\rho L)(\rho - v)^2 = 12.25$. Figure 2 displays a graph of R_L vs. b for the present example.

It is useful to note that the rent-maximizing condition $b^*L = Q_v/2$ is also the solution to the monopolist's problem: it pays to restrict production when $b^*L > Q_v/2$.

Figure 2. Simple rent vs. productivity b increase net revenues over the competitive condition.

Proof: A monopolist would seek to maximize $\pi_m = Q(P - v)$ subject to the production feasibility constraint. The output which solves the unconstrained problem is $Q_m = Q_v/2 = Q^*$. If this output is larger than bL, the monopolist cannot gain by restricting production because the land constraint is even more restrictive. The condition $bL < Q_v/2$ is also that defining b^*. Q.E.D.

Appropriability issues provide strategic interest to this simple situation. Suppose farmer A discovers a method for increasing productivity from 1.5 to $b^* = 1.75$ bundles per acre. If his plot of land is small and his innovation is not imitable, then Theorem 1 and the ensuing discussion of appropriability apply. However, if the innovation is fully imitable, Theorem 3 reveals that the innovation increases land rent, in this case from 12 to 12.25. No tenant farmer appropriates any value, but all land becomes more valuable. Had farmer A the chance, he could have amplified his gain from his imitable innovation by buying land (as much as possible).

After farmer A has bought land and shared his know-how, suppose farmer B discovers a second fully imitable productivity-enhancing innovation such that b increases from 1.75 to 2.5 . Because $b > b^*$, this increase in productivity actually decreases net revenue and, therefore, the value of all sunflower land. Now imitability hurts farmer B absolutely, because she cannot gain by owning or buying land beforehand. Farmer B can gain only if the innovation is nonimitable and her plot of land is small. Consumers are better off, but farmers and landowners are damaged by the higher productivity.

The idea that increases in productivity can diminish the value of scarce resources used in production seems counterintuitive to many. It is much more general than the model herein explored. This kind of effect was first pointed out by Nelson (1957), who called it a 'paradox of value theory.' His primary result, that land rent can increase with an increase in wage cost, is not present in our

[13] The result of Theorem 2 is not specific to this model; it can be obtained for more complex production functions and demand specifications, as long as there is a limited fixed factor in the industry.

model with homogeneous land. However, by introducing heterogeneous land we do obtain his result (see Theorem 4).

Thus far we have maintained that a sufficiently small farm can appropriate the gains to increased productivity. It is possible to be much more exact about the fractions of land and productivity increases that enable private appropriation to innovations. Suppose that a fraction $\gamma, 0 < \gamma \leq 1$, of the land is 'special' land. Specifically, an innovation leads to an increase in productivity on special land from b to βb, with $\beta > 1$, but this innovation does not result in increased productivity on the nonspecial land. The total output of all land is $(1 + \gamma(\beta - 1))bL = gbL$, where $g \equiv 1 + \gamma(\beta - 1)$. For there to be an economic scarcity of land, we require $gb < Q_v/L$.

> Theorem 4: Given $gb < Q_v/L$, there is a $\gamma^* > 0$ such that parcels of special land smaller than $\gamma^* L$ show increases in net revenues with increases in β. There is also a $\beta^* > 0$ such that for any parcel of land of size γL, if $\beta > \beta^*$, increases in β decrease the net revenues to the special land.

Proof: Net revenues to the special land are $R_\gamma = \gamma\beta bL(P - v)$. The derivative of R_γ with respect to β is $dR_\gamma/d\beta = \gamma bL[P - v + \beta d\mathrm{P}/d\beta]$. Working with $P = \rho\,(1 - gbL/Q_0)$, we find that $dR_\gamma/d\beta = 0$ implies $\gamma^* \equiv (Q_v/bL - 1)\,/(2\beta - 1)$. The numerator is positive because $gb < Q_v/L$, and the denominator is also positive because $\beta > 1$. The condition $dR_\gamma/d\beta = 0$ is also induced by $\beta = \beta^* \equiv [Q_v/bL - 1 + \gamma]\,/2\gamma$. Clearly, $gbL < Q_v$ implies $\beta^* > 0$. Q.E.D.

Continuing to work our example, set $\gamma = 0.2$ and $\beta = 2$. With 20 percent of the land 2 times as productive, $g = 1.2$: total production is up by 20 percent. The increased production pushes the market price of sunflowers down to 12.8 . The rent on the original land drops from 12 to 10.2 per acre because of the fall in the price of sunflowers. The rent on the good land is 20.4 per acre, $\beta = 2$ times as great as the poorer land. Increasing the productivity of the special land can generate large returns to the owner (but eventually the same limitations apply as we found for the homogeneous land). Setting $\gamma = 0.2, \beta^* = 23/6 = 3.83$, yielding a maximum attainable net revenue per acre of 26.45.

If $\beta = 7/6, \gamma^* = 1$. However, if $\beta = 2$, the largest plot supporting productivity innovation is $\gamma^* = 4/9 = 44$ percent of the total land.

Heterogeneous Land and Alternate Use

We now introduce into the model two kinds of heterogeneity: differing levels of variable cost and an alternate use for the land. We divide the 100 units of land into four equally sized parcels labeled A, B, C, and D. Each parcel has a different level of productivity and variable cost in sunflower production. In addition, the alternative use for parcels B, C, and D is the production of corn. Parcel A has no alternative use to sunflower production. Table 1 displays the parameters and

results for all four types of land. The price of corn is assumed to be unaffected by whether or not these parcels participate in corn production. [14]

Figure 3 displays the sunflower industry supply curve. Note that although the parcels are equal in acreage, they differ in total production capability. Parcel A land is drawn into production when the price of sunflowers rises above its variable cost of 9.5 per bundle. Whereas parcel B land has a lower [14] Were the price of corn to vary with the entry and exit of these parcels, the concept of the 'foregone net revenues in corn' would be rendered impotent. The simple rent 'foregone' in corn would vary with each acre's decision to enter or exit corn production. In such a setting, equilibrium prices and outputs solve a set of simultaneous conditions and the concept of opportunity cost has no traction. Much of the verbal paraphernalia of the neoclassical theory of the firm vanishes in a general equilibrium setting. variable cost in sunflowers (7.5), it will only come into sunflower production when the price rises to $4.5 + 7.5 = 12$ per bundle. Similar considerations for the remaining parcels flesh out the supply curve. Intersection with the demand curve provides a market-clearing price of 14.5 per bundle with 91.67 percent of the land being used for sunflower production and the remainder producing corn.

For each parcel of land, the price of 14.5 can be broken into three types of payment: (a) payments for wage costs, (b) payments to the land for foregone corn production (forgone simple rents in corn), and (c) payments to the land, over and above its alternative use value, for its extra value in sunflower production. The first is a standard cost concept, the second is the opportunity cost of the land, and the third is economic rent. The sum of (b) and (c) is simple rent. Note that parcel D earns no economic rent-it is marginal land to the sunflower industry. Parcel D is not marginal because it is inefficient, it is marginal because it teeters between corn and sunflower production. The slightest decrease in the price of sunflowers will push this whole parcel over to corn production.

The rents earned in the sunflower industry are not Ricardian rents. Many authors casually call the profits of a firm 'Ricardian rents' when they

wish to indicate that scarcity, rather than market power, creates the profit. This usage is unfortunate because the Ricardian concept neither enables rent to be measured by a resource's alternative productive uses, nor allows the comparison of rents across resources that have differing efficiencies in alternative uses. Writing to this issue, Alchian carefully defines Ricardian rents as existing only with respect to equal values in alternative use:

The rents accruing to different units of some otherwise homogeneous resource may differ and result in differences of rent over the next most valued use, differences that are called 'Ricardian rents.' This occurs when the factors, all regarded as of the same 'type' in other uses, are actually different with respect to some significant factor for its use here, though this factor, which is pertinent here, is irrelevant in any other uses. . . . For some questions, the inaccurate 'homogenization' can be a convenient simplification, but for explaining each unit's actual rents it can lead to confusions and misunderstanding. (Alchian, 1987: 142)

In our example, consider parcels A and B. The simple rents per bundle to these parcels are 5

Table 1. Parameters and accounts for four parcels

		A	B	C	D
Corn Farm Accounts					
a	Price Corn/Bushel		10.00	10.00	10.00
b	Wage Cost in Corn/Bushel		8.50	9.00	9.50
c = a − b	Simple Rent in Corn/Bushel		1.50	1.00	0.50
d = c/a	Simple Rent Margin in Corn		15%	10%	5%
e	Productivity: Corn Bushels/Acre		3.00	3.00	3.00
f = e ∗ c	Simple Rent in Corn/Acre		4.50	3.00	1.50
Sunflower Farm Accounts					
g: from Sup = Dmd	Price of Sunflowers/Bundle	14.50	14.50	14.50	14.50
h	Wage Cost in Sunflowers/Bundle	9.50	7.50	11.50	13.50
j = g − h	Simple Rent in Sunflowers/Bundle	5.00	7.00	3.00	1.00
k = j/g	Simple Rent Margin in Sunflowers	34%	48%	21%	7%
m	Productivity: Sunflowers Bundles/Acre	2.00	1.00	1.50	1.50
n = j ∗ m	Simple Rent in Sunflowers/Acre	10.00	7.00	4.50	1.50
p = n − f	Economic Rent = SR in Sunflowers − SR in Corn		2.50	1.50	0.00
Sunflower Supply Curve					
q = f/m	Foregone Simple Corn Rent/Sunflowers Bundle	-	4.50	2.00	1.00
r = q + h	Supply Cost of Sunflowers/Bundle	9.50	12.00	13.50	14.50
s = 25 ∗ m	Potential Parcel Production in Sunflowers	50.00	25.00	37.50	37.50
t = sum(s)	Cumulative Potential Sunflower Production	50.00	75.00	112.50	150.00
u = D(R)	Demand at Supply Cost	262.50	200.00	162.50	137.50

and 7 respectively. But the economic rents per bundle are 5 and 2.5 respectively. The difference arises because parcel B has a value in alternative use that corresponds to 4.5 per bundle.

Simple rent measures the value of the scarce resource. The payments for land paid by tenant farmers to landlords are simple rent. These payments are also the net cash flows that would be discounted to value the land. Were the resource not scare ($bL > Q_v$), then simple rent would be zero.

Economic rent is unobservable. Economic rent is a portion of simple rent. Neoclassical theory divides the simple rent payment into pieces: that part which is paid to the land to compensate it for 'not growing corn' and that part which is paid to the land for its scarcity value in sunflowers. The former is called an opportunity cost or value in next-best use; the latter is called an economic rent. It is important to understand that economic rent is not observable in the normal performance of real firms. It only is revealed by decisions to move resources from one industry or one use to another.

The market price of a resource is simple rent. Economic rent, by contrast, is the difference between the market price of a resource and its opportunity cost; the later is a value which is a conjecture about a price in a different state of the world. The whole structure of partial equilibrium analysis rests on the construction of such 'costs' and 'cost functions' which are market prices only at the one point of equilibrium where supply equals demand.

Consider the economic rent associated with Dell Computer's supply-chain skills. It is the differ-

ence between the payment for these skills less the payment they would receive in their 'next-best use.' But 'next-best use' has no operational meaning in this context. It has a clear meaning only in an economy consisting of N traded commodities (inputs and outputs). So, although it is clear that these skills are 'scarce,' and it is also clear that they have a myriad of alternative uses, the concept of economic rent has little bite in this context. We address this issue in more detail in the section on Economic Cost.

The payments received by the land would be unchanged were it inalienable. Suppose that the land in our model were owned by individual (small) farmers. Suppose further that these farmers can not sell their land: it is not alienable. In addition, assume that the farmers are rational maximizers and that farmers do not possess land-specific or crop-specific skills. Under such assumptions, the land uses, prices, and simple rent payments received are unchanged from those we have computed. The only difference is that the farmer cannot sell his right to the cash flows generated by the land. [15]

Were the land unpriced, the simple rents would be termed profits. Resources are unpriced when property rights are difficult to define, and where the transfer of use-rights is technically or economically difficult. Although it is hard to imagine unpriced land, many resources do exist that are limited in extent and heterogeneous in quality across firms. Consider, for example, gate locations at airports or cumulative tacit knowledge. The key characteristic of the resource in question is that it is economically scarce: when demand for it increases, the result is a higher rent payment (or higher value) rather than increased supply. Were the land in this example unpriced, [16] the returns to it would be named 'profits' rather than simple rents, and the profit margins of each parcel would be the simple rent margins shown in Table 1.

Were these lands unpriced, how would analysts appraise the performance of the four parcels? In Table 1, parcel B has the highest profitability, the most common measure of performance. On the other hand, parcel A has the highest land rent or value per acre. The difference is merely the result of scaling: the difference is not any more meaningful than saying aluminum is worth more per ounce than lead. The comparison depends upon the scaling factor. Lead is worth more per cubic inch than aluminum. [17] We note, importantly, that there is no known scaling for knowledge, knowhow, or reputation.

Measures of performance in use today, such as profit margin and return on investment, are no better than the simple rent margins in Table 1. Such ratios measure the relative importance of unpriced vs. priced resources. Tobin's q, for example, is the ratio of the value of all of the firm's resources to an estimate of the market value of its priced resources.

To understand the problem with these measures, turn the logic around: would a firm with a high ROI or high T_q be 'lower' performing if all of its resources were properly priced? Were the four

[15] The Fisher-Hirshleifer separation theorem holds that, if capital markets are perfect, a rational farmer would be indifferent between selling the land and receiving the cash flow. That is, whether to take the cash flows or their net present value is a consumption timing choice, not an investment decision. The timing of consumption of wealth can be arranged by borrowing and lending: it does not require the sale of productive assets.

parcels of 'unpriced' land suddenly priced, none of their cash flows would change. But each would then be seen to simply earn a 6 percent return on market value.

Wage Cost Innovation

In the section on Innovation and Appropriation we dealt with the issue of productivity innovation. With homogeneous land, increases in wage cost v unequivocally reduced all simple rents. This occurs because farm output does not fall when v is marginally increased, so there is no consequent upward pressure on price. In the four-parcel situation, however, Nelson's 'paradox' is present.

Recall that our 'wage costs' include the costs of seeds, supplies, water, labor, etc. We introduce a factor θ which multiplies all wage costs in sunflowers. It can be taken as measuring the relative level of sunflower seed prices. For the parameters shown in Table $1, \theta = 1$.

In the starting equilibrium, parcels A, B, and C are fully devoted to sunflowers and parcel D is 2/3 in sunflowers and 1/3 in corn. If θ is increased slightly above 1 , raising wage costs for all parcels, some of parcel D's acres move to corn, increasing the price of sunflowers. Parcel D's net revenue per acre is locked to that in corn because it responds to changes by moving acres between sunflowers and corn. But, for parcels A, B, and C, there are two conflicting effects: both revenue and wage cost per bundle rise. In our example, the increase in wage cost does actually increase net revenues for these parcels. Higher costs can lead to higher returns to scarce resources!

Figure 4 shows the simple rent per acre for each parcel and for all parcels together (in best use) as a function of θ. As shown, simple rents increase with θ until $\theta = 1.074$, the point at which 100 percent of parcel D is devoted to corn. Further increases in θ reduce rents for all remaining parcels until parcel C begins to shift acres into corn. At that point, parcel C's rents are fixed and parcels A and

B again show increases in rent with higher costs, until parcel C is entirely devoted to corn.

Firm Cost and Industry Cost

The tension between conflicting concepts in the neoclassical treatment of rent is clearest in the construction of the industry supply function. In this section we show that despite the neoclassical claim that the industry supply function is the horizontal sum of firm supply curves, this is not the case in the sunflower industry, or any industry using an economically scarce resource. The problem is inherent in partial equilibrium analysis. One of its consequences, more fully explored in the section on Economic Cost, is the continuing confusion over the meaning of 'cost.'

[16] The issue is not tangibility but pricing. There can be intangibles (intellectual property) that are priced and tangibles (Intel's most modern 300 mm wafer processing fab) that are not. Note also that there can be property that is inalienable, yet 'priced' because it is comparable to other priced properties. [17] As we write, aluminum is 3.2 times the price of lead per pound, and lead is 1.3 times the price of aluminum per cubic inch.

Every microeconomics text shows how to construct an industry supply curve: it is the horizontal sum of the individual firm supply curves. That is, if the quantity produced by the i th firm at industry price P is $S_i(P)$, then the industry supply curve is $S(P) = \sum_i S_i(P)$.[18] As Varian (2003: 397) puts it, 'the industry supply curve, or the market supply curve is...the sum of the individual supply curves. Geometrically, we take the sum of the quantities supplied by each firm at each price, which gives us a horizontal sum of the supply curves.'

It is also standard doctrine that in the 'long run' most factors are variable. Our story of land and farmers is a long-run story: wage-like inputs are variable and the amount of land is fixed forever. It is standard doctrine that this is the condition in which true rents, rather than quasi-rents, are induced. Consider a small farmer's cost function in the long run. Each small price-taking farmer faces a 'price' v for the wage cost of a bushel [19] and a price R_L for land. The long-run cost function for producing q bundles of sunflowers is $C(q) = vq + (R_L/b)\,q$. The firm's long-run marginal cost function is $C_q = v + R_L/b$. Plugging in values for parcel A, we find that $C_q = 9.5 + 10/2 = 14.5$, which is independent of q and exactly equal to the price. It should not be surprising that $C_q = P$ because we defined $R_L = b(P - v)$ in the first place!

Applying this logic to each parcel, each has a long-run supply function that is a horizontal line at 14.5 (because the rent is a result of price, rent per bushel differs from parcel to parcel by just enough to cancel out the differences in wage cost). Their sum across farms and parcels is the same horizontal line. This naturally intersects the demand curve at a price of 14.5 and magically produces the equilibrium amount of land in use. But it is a contrived result; the reasoning is circular. The 'cost' of land was computed backwards from knowledge of the price.

To see the impact of this circular reasoning, look at what happens if we move the demand function down. In particular, let ρ drop from 20 to 18 (Figure 5). The new price is not the intersection of the shifted demand function and the old long-run supply function. If it were, the new price would have to be 14.5 again. The new price is 13.95 and parcel D drops out of sunflowers, returning to corn production. Simple rent per acre in parcel A drops from 10 to 8.9. Instead of determining price, the long-run average cost curve falls to follow the price. Whereas it had been a horizontal line at 14.5, after the change in demand the supply curve becomes a horizontal line at $v + R_L/b = 9.5 + 8.9/2 = 13.95$, the new price. The famous 'scissors' of Marshall is lost-the two blades do not cut independently. Instead, the long-run 'supply curve' shifts up and down right along with the demand curve. The equilibrium price cannot be determined as an intersection of the demand curve and the long-run supply curve.

Instead, the price is discovered as the intersection of the demand curve and the long-run average

[18] The supply curve is the inverse of the marginal cost curve. That is, if $C(q)$ is the function relating output q to cost, then $C_q = dC/dq$ is the marginal cost curve and its inverse is defined by $S_i\left(C_q(q)\right) = q$. If the mapping is multivalued, only the value for $C_{qq} > 0$ is chosen. Furthermore, $S_i(q) = 0$ if $\pi_i(q) < 0$ (the firm does not produce unless profits are non-negative). [19] Technically, $v = \mathbf{x}'\mathbf{z}$, where $\mathbf{x}$ is the vector of inputs required per bushel produced and $\mathbf{z}$ is the vector of their prices per unit of input. We simplify the exposition by working with the inner product, the price per bushel of inputs.

cost function absent economic rent paid on resources used within the industry. That is, the operative long-run average supply function contains wage costs plus the simple rents that would be paid in corn, but not the simple rents that are paid in sunflowers. Neoclassical theory has two ways of explaining this odd work-around. The first is to simply state a rule: when computing the supply function, leave out the rents.

This is the approach taken by Nobel Laureate Gary Becker: [20]

> The difference between the revenue and cost of an efficient firm is called 'economic rent,' and is a reward for the scarcity of superior efficiency that would vanish as it became common. ... If the superior resources and abilities responsible for economic rent can be sold to other firms, as can be with a mine or a patent right, economic rent might justifiably be called a 'cost.' If it were added to other costs, rent-inclusive total cost would equal total revenue at the equilibrium position and exceed revenue at all other positions. ... This treatment of rent satisfies the accountant's desire to make both sides of the ledger balance, but has several shortcomings. The most important is that rents are price-determined, not price-determining, as are labor, capital, and other conventional costs. ... Rent is price-determined because an upward shift in the demand curve would increase the equilibrium price in the industry if the industry supply curve were positively inclined and thus increase the rent of all firms already in the industry. . . . Since the purpose of cost curves is to help determine equilibrium quantity and price, only price-determining costs should be included in these curves. (Becker, 1971: 77)

Becker's argument is that if one calls a rent a 'cost' a problem arises in that this 'cost' doesn't enter into the supply curve. True, there is a problem, but it isn't, as he suggests, with 'accountant's desires.' Rather, it is with the concept of cost in partial equilibrium analysis. In the above argument, Becker justifies his approach by reference to firm-specific rents. [21] But it is also necessary, as we have seen for the sunflower industry, to 'remove' rents due to industry-specific resources, or even resources presently used in sunflowers that could be used in corn. This procedure drives a wedge between the theory of the firm and the theory of price-setting. That is, what is a cost to the firm (rent on land) disappears when the problem becomes determining price. This kind of break between private and social costs is called an externality. It is disconcerting to find a big externality right in the center of a perfectly competitive model.

The second approach to this conundrum is to name it a 'pecuniary externality' or a 'factor-price effect.' Hirshleifer (1980: 314) uses both terms, noting that 'For a "pecuniary" economy or

[20] Note that Becker's 'economic rent' is a rent payment to a scarce resource. It is only a true 'economic rent' if the resource has no alternative use. Becker's usage follows that of many other economists: it is assumed that unpriced resources are specialized, firm-specific, and have no alternative use value. Of course, if resources have no alternative use values, there is no problem of strategy because nothing else can be done except that which is being done. [21] Note that although Becker claims these are rewards to superior efficiency, 'economic cost' is defined as value in alternate use, so that a firm that is efficient at both its focal and a neighboring industry earns little 'economic rent.' There is no particular reason to believe that entrepreneurial skill is more specific to an industry than are tangible investments in plant and equipment.

diseconomy, the interaction between industry output and firm cost function comes about solely through changes in the market prices of inputs.' As can be seen from the sunflower industry, the 'factor-price effect' is not just a slight adjustment; it is fundamental. All firm-specific and industry-specific rents (costs) must be totally removed from the supply curve. After doing that, additional adjustments must be made for any increase in rents to scarce factors in other industries caused by the focal industry's demand for inputs. All of these 'adjustments' are post hoc patches to partial equilibrium analysis.

Our example makes clear that price-taking small firms are not sufficient to support the neo-classical supply function. The additional assumption necessary to enable a stable long-run supply function is that the entire industry be 'small' with regard to each resource pool it employs.

The divorce between firm and industry costs induced by rents has been a source of confusion and concern since the introduction, in the late nineteenth century, of partial equilibrium analysis and the opportunity cost concept. The divorce is generally ignored or swept under the rug because the pecuniary externality 'does no harm.' That is, the pecuniary externality is not, like a technological externality, a cost to society. Instead, it is an error in partial equilibrium analysis that vanishes in a general equilibrium framework. Addressing the source of this problem, Shubik (1971: 713) said, 'The apparent externality shows up as a feedback from one market to another in a comparative statics explanation of adjustment to change. The paradox of the pecuniary externality is caused by the nature of the partial equilibrium analysis of the firm.'

In the case of sunflower production, the pecuniary externality arises because the 'cost' to a single farmer of producing one more bundle of sunflowers, $v + R_L/b$, cannot be aggregated across firms. The industry as a whole cannot produce more sunflowers. The single small farmer produces more sunflowers by buying or renting more land, but the land he acquires was already producing sunflowers. The cost concept breaks because the so-called 'cost' of the land to the farmer is not a 'cost' to the industry or society-the land acquired was not in an alternate or next-best use; it was already in its best use.

Economic Rent, Simple Rent, and Rent Sensitivity Analysis

Economic rent demarks scarcity, simple rent determines value. Given a durable resource, the simple rent is the total payment received by the owner of that resource. The value of the resource is the discounted value of future simple rent payments. The economic rent is a portion of the simple rent and depends upon the analyst's definition of 'alternative use.' Given a specified alternative use, economic rent is the difference between the present simple rent payment and the simple rent payment that would be received in that alternative use. Such alternative payments are out-of-equilibrium conjectures based upon various assumptions about what might change (and what might not) to force the resource into its alternative use.

The presence of economic rent demarks economic scarcity because, absent scarcity, there would

be enough to supply both uses and, consequently, the two use values would be the same. For example, sunflower land is scarce whereas another durable resource, say a pickup truck, is not. Both receive simple rent payments in equilibrium. The difference between the two resources, in neoclassical terms, is their value in alternative use. Parcel A land has no value outside the sunflower industry, whereas the pickup truck is valued equally across many industries. Its value is set by its cost of manufacture because it is in elastic supply to the whole range of industries, being neither scarce nor specialized.

The concept of economic rent is critical in distinguishing between scarce and nonscarce resources. However, it relies on a fairly arbitrary division of all activities and decisions into 'this use' and 'alternative uses.' We proffer another way of identifying scarce resources that is closely related to the economic rent concept but which comes at the question from a different direction. We call this approach rent sensitivity analysis. The basic idea is that instead of calculating the amount of economic rent, one can examine the sensitivity of simple rent to changes in demand.

Consider an increase in demand for sunflowers as represented by an increase in the chokeprice ρ. If land is scarce, then from Equation 3 we have $\mathrm{d}R_L/\mathrm{d}\rho = b\,(1 - bL/Q_0) > 0$. If land were not scarce, then $P_L = v, R_L = 0$, and $dR_L/d\rho = 0$. Thus, the payment to land is sensitive to changes in demand if and only if land is economically scarce and if there are economic rents. By contrast, the payment for a pickup truck does not vary with the demand for sunflowers.

Rent sensitivity analysis provides a simple integrated way of looking at resource scarcity and value. If the demand for personal computers rises, which resources become more valuable? If the demand for laptop computers rises whereas the demand for desktops falls, which resources become more valuable? If the demand for Dell brand computers rises, which resource becomes more valuable? Intuition about these questions is generally stronger than about the value of the resource in alternative uses.

Once the framework for rent sensitivity analysis has been established, it should be clear that resource values depend upon any number of parameters other than demand. The very meaning of specialization is that a resource's value is especially sensitive to certain parameters. This approach can be used to discover the sensitivity of resource values to changes in demand, changes in the prices of various inputs, changes in law, etc.

Economic Cost

In the sunflower industry, it is easy to see that land is seeking its highest-valued use. It is much more difficult to understand the meaning of the phrase 'the cost of land'. On the one hand, land appears to be a cost to the farmer, for surely each farmer must pay for land. On the other hand, it is the sole repository of value in the story, the act of farming being profitless. Furthermore, the rental rate of land is not actually the 'cost' of land used to construct the supply curve. There, only alternative out-of-equilibrium values are used. Finally, the 'market price' or 'cost' of land swings up

and down directly with the demand for sunflowers, behaving more like a 'profit' than a cost. What is going on here? In this section we explore various views on the meaning of cost and provide our own perspective.

Opportunity Cost

The history of economic cost concepts is complex. The classical writers had a simple physical or labor theory of cost. For Smith, because a hunter could catch three beavers or one bear in a day's work, the cost of a bear was three beavers. For Marshall, cost was more complex, but was ultimately drawn from the pain or sacrifice required to do something.

Wieser (1893) first articulated the Austrian school's position that relative prices measure the relative values of foregone opportunities. This position was derived from the analysis of consumer indifference curves. The assertion found in modern economic textbooks is that economic cost is opportunity cost. This definition produces little debate. There is, unfortunately, less certainty about the definition of opportunity cost. [22]

Standard Textbook Doctrine

It is commonplace in textbooks on microeconomics to assert that opportunity cost takes into account alternative use value. Cohen and Cyert (1965: 92) say, 'the cost of using a productive factor to produce any commodity is the value of the best opportunity which is foregone by not using this factor in another way.' Besanko, Dranove, and Shanley (2000: 20) are quite specific that the cost of using a resource in one way is the value it could have obtained in its next-best use: 'the economic cost of deploying resources in a particular activity is the value of the best foregone alternative use of these resources.'

Nobel Laureate Paul Samuelson's textbook (17th edition, now co-authored with Nordhaus) emphasizes the tight connection between opportunity costs and resource prices:

Economic cost includes not only the obvious out-of-pocket purchases or monetary transactions but also more subtle opportunity costs, such as the return to labor supplied by the owner of a firm. These opportunity costs are tightly constrained by the bids and offers in competitive markets, so price is close to opportunity cost for marketed goods and services. (Samuelson and Nordhaus, 2001: Ch. 7)

As we have already seen in sunflowers, the alternative use value of the land, used to build a supply curve, is not its market price. This divergence is rarely addressed.

[22] It is worth noting that the authors could find no reference to opportunity cost or to rent in advanced microeconomics or general equilibrium texts. These books simply introduce the correspondences between production and cost functions and get on with the mathematics.

Nobel Laureate Joseph Stiglitz's microeconomics textbook does not mention alternative use value, but does include the cost of capital:

> It is important that cost include all opportunity costs of production borne by the firm. Not only are such explicit costs as wages, energy, raw materials, and interest included, but also more subtle opportunity costs are taken into account, such as the value of the entrepreneur's time or the alternative earnings on the equity invested by the owners of the firm. These are considered opportunity costs because if the firm did not produce, the entrepreneur would devote time to some other activity, and the owners would take their investment capital elsewhere. (Stiglitz, 1997: Ch. 12)

Taking the expected return on equity investment as a cost is, in our view, a mistake. (Many authors do not make this inclusion.) The technical issue is whether or not the capital has been irreversibly specialized to a use, in which case it cannot be redeployed and may have an 'opportunity cost' of zero. We expand on this point in the subsection on EVA and the Expected Return Fallacy.

The opportunity cost of a resource is its value in its next-best use. Despite the apparent precision of the word 'best,' there remains ambiguity as to what is meant by 'alternative use.' In looking at alternative uses, how close to the current use are we allowed to look? To put the 'use' issue more precisely, return again to the sunflower industry and focus on parcel B. Its simple rent is 7 per acre, but it could get 4.5 per acre in corn. Building a supply curve is an exercise in tracing out the amount supplied at various out-of-equilibrium prices. For parcel B, its contribution to the supply curve is the second chunk (Figure 3), which tells us that it switches from corn to sunflowers at a price of 12 or higher. But the equilibrium price is 14.5 , and parcel B is fully in sunflowers. In equilibrium, is the 4.5 it could get in corn still a 'cost' to parcel B farmers? Were this a real farm, and not a mathematical model, the 'next-best' alternative might actually be to use a slightly less expensive grade of fertilizer, yielding fewer bundles per acre, garnering a simple rent of say 6.9 instead of 7 .

That option would have as great a claim to being the farm's opportunity cost as does the less attractive option of growing corn. If one argues that the 'poorer' fertilizer option shouldn't be considered because it is 'not optimal,' why count the nonoptimal corn alternative, which is even more foolish? If one argues that a drop in sunflower prices will indeed make growing corn optimal, why not also argue that a drop in fertilizer prices will also make its use optimal? [23]

Stigler'S 'Point of View' Doctrine

Nobel Laureate George Stigler (1966: 105) tries to explain the alternative use doctrine this way:

The alternative uses of a resource depend upon the use for which cost is being reckoned:

1. The cost of an acre of land to agricultural uses is the amount the land could yield in nonagricultural uses (residences, parks, and so on).

2. The cost of an acre of land to the wheat-growing industry is the amount it would yield in other agricultural crops (oats, corn, and so on), as well as nonagricultural uses.

3. The cost of an acre of land to wheat farmer X is the amount the land could yield to other wheat farmers, as well as non-wheat uses.

Thus, according to Stigler, the cost of land to the farmer is not the cost of land to the industry, which is, in turn, not the cost of land to society. This doctrine addresses the question of specifying alternative use by reference to three levels of aggregation: the firm, the industry, and society. It does not attempt to address the question of why these are the relevant alternative use settings. A major problem with this doctrine is that it abandons the most important feature of the whole cost concept: cost is borne by a decision-maker.

The Coase-buchanan View: Cost and Choice

Whereas Stigler ties cost to use, other economists tie cost to choice. This line of reasoning is best exemplified by Nobel Laureate James Buchanan, writing the 'opportunity cost' entry in the New Palgrave: Scarcity introduces the necessity of choice. ... Choice implies rejected as well as selected alternatives. Opportunity cost is the evaluation placed on the most highly valued of the rejected alternatives or opportunities. It is that value that is given up or sacrificed in order to secure the higher value that selection of the chosen object embodies. . . . a chooser is required, a person who decides ... The value placed on the option that is not chosen, the opportunity cost, must be that value that exists in the mind of the individual who chooses. (Buchanan, 1987: 719)

Nobel Laureate Ronald Coase, a careful thinker about the junction between economic theory and business practice, agreed with Buchanan and makes it even clearer: only that single action which covers costs is the best possible action:

> The cost of doing anything consists of the receipts which could have been obtained if that particular decision had not been taken. . . . Costs will only be covered if he chooses, out of the various courses of action which seem open to him, that one which maximizes his profits. To cover costs and to maximize profits are essentially two ways of expressing the same phenomenon. (Coase, 1938: 123)

In the same vein, Buchanan (1969: 28) took the argument one step further, writing 'Any profit opportunity that is within the realm of possibility but which is rejected becomes a cost of undertaking the preferred course of action.'

[23] The 'answer' is that partial equilibrium analysis is about determining the industry price of sunflowers, assuming that the prices of corn, fertilizer, etc., are unchanged.

The Coase-Buchanan view leads directly to the conclusion that Cost = Revenue. The reasoning is simple: for any possible difference between value and cost, a finer look at the problem ('hairsplitting') will always reveal a possible action or use that is closer to the optimal. In the limit, Cost = Revenue.

The Coase-Buchanan position is derived from what has become known as the subjectivist, or London school, which, in turn, traces its origins to Wicksteed (1910) and especially Robbins (1932). It is this school of thought which taught that sunk cost does not matter: economic cost is a concept that applies to forward-looking decisions, not to what has happened in the past. About the London opportunity cost doctrine, Nobel Laureate Paul Samuelson (1974: 234, fn. 32) once said: 'the usual formulation is rather a mumbo jumbo of high sounding gibberish, which does not state the conditions of equilibrium in a very direct way.'

Buchanan was quite aware that the concept of opportunity cost implies Cost = Revenue . He wrote:

Despite the necessity of accepting this straightforward result [cost equals revenue] of apparently consistent opportunity-cost reasoning, economists were-and are-extremely reluctant to take this step. To include all foregone profits as costs plays havoc with the whole cost-curve apparatus that is part of our stock-in-trade. And without this how could we teach elementary price theory? (Buchanan, 1969: 28)

Friedman'S Noncontractual Costs

Nobel Laureate Milton Friedman writes on price theory in a way that avoids most of the language of opportunity cost. Nevertheless, he arrives at the same place as Buchanan and Coase. He simply defines cost as equal to revenue and guarantees the equality by labeling a category of cost (noncontractual cost, equivalent to an entrepreneurial fee) that makes up any difference.

> All in all, the best procedure seems to me to be to define total cost as identical with total receiptsto make the totals of two sides of a double-entry account. One can then distinguish between different kinds of costs, the chief distinction in pure theory being between costs that depend on what the firm does but not on how its actions turn out (contractual costs) and the rest of its costs or receipts (noncontractual costs). The former represent the cost of factors of production viewed solely as 'hired' resources capable of being rented out to other firms; the latter represent payment for whatever it is that makes identical collections of resources different when employed by different firms-a factor of production that we may formally designate entrepreneurial capacity, recognizing that this term gives a name to our ignorance rather than dispelling it. (Friedman, 1976: 148-149, underline emphasis added)

Friedman's bold position on cost seems to resolve many ambiguities. Friedman's recommen-

dation that 'it seems best' to define Cost = Revenue draws on the principle of full imputation. As Winter (1987: 165) reminds us, 'This principle states that a proper economic valuation of a collection of resources is one that precisely accounts for the returns the resources make possible.'

If Cost ≡ Revenue, then dividing both sides of the identity by output yields Average Cost ≡ Price. Addressing the second identity, Friedman hesitates:

> The existence of specialized resources ... makes it impossible to define the average cost of a particular firm for different hypothetical outputs independently of conditions of demand. The returns to the specialized factors are now 'rents,' at least in part, and, in consequence, do not determine the price, but are determined by it. . . . The equality between price and average cost, in any sense in which it is more than a truism, thus reflects competition in the capital market and has no relation to the state of competition in product or factor markets. (Friedman,1976: 147-148, emphasis added)

Like Becker (see quote on page 916), Friedman recognizes all too well that Average Cost ≡ Price makes a shambles of long-run partial equilibrium analysis. In particular, the long-run marginal cost curves of firms do not aggregate to determine price. Instead, price determines the location of the long-run cost and supply curves.

The solution to this conundrum is to recognize that price is an outcome determined by general equilibrium (or bargaining). Given price, we push Friedman's analysis to its logical conclusion: factors are valued either in factor markets, or in capital markets, or by the opportunities to borrow against their payment streams. The essential difference between what he calls 'specialized factors' and 'hired factors' is that the former are not priced in factor markets. Were they priced, those prices would be automatically labeled 'costs.' The authors are less comfortable with the appellation 'specialized,' preferring the description 'scarce unpriced' factors. There is no particular reason to believe that firms' rent-producing scarce unpriced factors are more specialized than, for example, their plant and equipment. Resource values derive from scarcity, not specialization.

Unpriced Factors

For resource-based theory, the heart of the confusion over rents and cost lies in the treatment of unpriced factors. The payments (returns) to an unpriced factor are called 'profits' because the factor is unpriced; were it priced, economists would call the payments 'costs.' This fact has led strategy scholars to the incorrect position that unpriced factors are a source of competitive advantage. If a resource is a source of competitive advantage, it is because it is a superior resource, not because it is unpriced. Labeling the payment to an unpriced resource a 'profit' rather than a 'cost' is an error in the micro-foundations of neoclassical theory, not an indicator of value.

To more clearly understand this problem in micro-foundations, consider Caves' (1980: 65) well-known remark: 'At least some [factors] are simply not traded on open markets that permit

capitalizing their differential qualities into their contract prices. Thus rents that the firm can earn are not entirely passed along to the unique fixed factors responsible for them.'

For resource-based scholars, there are two problems with Caves' perspective. First, Caves' position is that factors cannot 'absorb' rents unless the factors are priced on open markets. This leaves a residual: firm profit. But firm profit is exactly the value of the 'unpriced' resources. There is no problem in valuation; the problem is in the language and concepts of the neoclassical position.

The second problem is that if all the rents were attributable to priced factors, they would be dismissed as simply 'factor returns.' Samuelson's textbook, now co-authored with Nordorf in its 17th edition, contains the following treatment of a supposedly 'profitable' firm. This treatment illustrates the 'factor absorption' logic at work (Samuelson and Nordorf, 2001: 447): 'If you own some special factors of production-like rich ore/land, exceptional know-how, or fertile soil-your accounts may show a high return even in the long run, but we realize that this is not so much profit as rent return to that special factor of production you are lucky enough to own.'

In short, the neoclassical position is that profits arise because some factors are not priced in markets; were these factors priced in markets, all of a firm's value would be dismissed as being due to the ownership of valuable factors. But factors can be valuable whether or not they are priced in the factor markets. Dell Computer owns valuable scarce factors that are not priced in factor markets: their value is expressed in the price of its common stock. And, because the company has issued public stock, the claims on those valuable factors are sold and resold every day. The fact that Dell's resources are not priced in factor markets, and the fact that they are valued and traded on a public capital market, does not diminish Dell's competitive advantage.

Quite a few resource-based scholars have internalized the standard economic position as exemplified in the above quotes from Samuelson and Caves. For example, Dierickx and Cool argue that the outside bid price for a resource is an opportunity cost, reducing the 'advantage' it can provide: sold in corresponding factor markets . . . However, the deployment of such assets does not entail a sustainable competitive advantage, precisely because they are freely tradeable ... if a privileged product market position is achieved or protected by the deployment of scarce assets, it is necessary to account for the opportunity costs of those assets. (Dierickx and Cool, 1989: 1505)

In another context, Peteraf (1993) defines 'imperfectly mobile resources' as those that are specialized to the firm. She notes that such resources 'can be a source of competitive advantage' because the 'rents generated by the asset will not be offset entirely by accounting for the asset's opportunity cost' (i.e., its value to others). [24]

These arguments use standard neoclassical reasoning. Our position is that they import logical errors from the neoclassical paradigm. The proper identity, in our view, is that net revenues are the sum of all payments to resources. The revenues of the firm should be allocated as payments to the inputs and resources that the firm employs. There is no residual, no unclaimed payment. Rich veins of ore, brand-names, and entrepreneurial skill can all receive payments for their services. All

Many inputs required for the implementation of a firm's product market strategy may be bought and

are 'factors, and that label does nothing to erase the value of the payment received. We rejoice, not lament, when competitors covet our resources. The enemies of value are imitation and substitution, not the bids of others.

Imagine two pharmaceutical firms, each earning $100 million in profit. Both have the same market capitalization. Firm X's profits are attributed to its portfolio of patents. Firm Y's profits are attributable to its organizational acumen and managerial skills. According to the Caves-Samuelson doctrine, firm X's economic profits are zero because they are due to valuable factors (patents) which could be sold or rented instead of used in the business. The opportunity cost of the resources is equal to the profit they generate. Firm Y, on the other hand, seems to have an economic profit because its valuable resources are not priced in factor markets. But are we to actually define competitive advantage in a way such that firm Y has it and firm X does not? What if a wholly new market mechanism made firm Y's skills marketable and priced; would that 'hurt' the firm or diminish its value? The problem here is not with the idea of advantage, but with the ideas of cost and profit! The truth is that economic profit is a chimera; the only source of wealth in the economy is the ownership of valuable factors. Owning a valuable resource is a good thing, and making its value higher is a good thing. And it is an even better thing if the resource is priced and saleable.

Full Imputation to Valuable Resources

Given the Buchanan-Coase position that cost follows choice, the only way to avoid the equality of cost with revenue would be to artificially restrict the domain of choice. That artificial restriction has been accomplished within neoclassical price theory by restricting the firm's choices of inputs and outputs to a set of N known commodities. The problems with the opportunity cost concept arise when neoclassical theory tries to use the theory of the firm to explore questions other than the determination of commodity prices. The alternative use constraints then vanish.

At least in Stigler's doctrine, the artificial restrictions on the choice set are explicit. He makes clear what many others do not: the concept of cost changes dramatically from one moment to another, depending upon the 'viewpoint.' The use of sudden, often hidden, changes in viewpoint is a tool of the trade in verbal economic discourse, especially with regard to opportunity costs. [25] We call a veiled shift in point of view a SPOV. Hidden SPOVs make it difficult to pin down what the neoclassical foundational elements actually are.

To illustrate SPOVs with regard to the opportunity cost concept, consider a firm hiring a temporary worker for one day for $300. The cost of this worker may be measured in at least six ways, depending upon the writer and the argument: (1) the money cost of the worker; (2) what the firm could have done with $300 spent in another way; (3) what the worker could have earned elsewhere

[24] In the same article, Peteraf correctly notes that under conditions of co-specialization 'the rents will be shared between the factor owners and the firm employing them. . . . This implies that the situation might be characterized best as bilateral monopoly, in which the distribution of rents is indeterminate.' The arguments in our companion paper expand upon the implications of resource interdependencies.

that day; (4) the value to the worker of what else could have been done with that day; (5) the monetary value of the product the worker could have produced working elsewhere; and (6) what the firm could have gained by employing the worker on a different task. SPOVs such as these mean that a key micro-foundation element is unclear in the realm of economics: the problems are compounded when the element is transferred to another field, such as to strategy's resource-based view.

The Buchanan-Coase argument is logically consistent. If cost is bound to a decision-maker's choices, one must adopt the point of view of the chooser, not suddenly switching between the past and present, the firm and the industry. In addition, once choice is the basis of cost, that which has been rejected fades away, and the standard of value-the hurdle to be overcome by any further decision-is the value presently obtained. The 'cost' of a resource is the net revenue it yields under best use, just as the 'cost' of an acre of farmland is what one has to pay to procure it.

This last point deserves further comment. Consider the textbook writers' favorite example of opportunity cost: an entrepreneur runs a firm which reports a profit of 100 that she takes as income. The economist notes that the entrepreneur has 'forgotten' to account for her own cost of time. In a regular job, she could earn 60, so the profits of the firm are 40 rather than 100 . What the economist's computation tells us is that it is better for her to run the firm than to abandon it for the regular job. That is the entire and complete content. The 60 foregone is no more a 'cost' to the firm than a sunk investment is a cost. To argue that 60 is the 'cost' of the entrepreneur, one has to tell a story about how she forewent the 60 to get the 100 . But that is history. There is no cost in the textbook story because there is no decision under consideration. Were the entrepreneur to consider spending her effort on some other activity, 100 is the number to beat, not 60 .

Another way of looking at the entrepreneur's problem is that the formal neoclassical definition of economic profit poses (by implication) the question 'Should this firm be started?' If the economic profit is negative, the firm should not be started. [26] If the answer is affirmative, there is a wedge between the outside and inside use values of the resources employed. In this case, any cost (price) lying between the outside and inside use value can be assigned to the resource without changing the decision to start the firm. In our companion paper we show that this indeterminacy appears in modern economics as the bargaining frontier.

In the story of the sunflower land, recall the curious behavior of the 'cost' of land. Rather than being fixed by input prices, the cost (price) of sunflower land varies directly with the demand for sunflowers. If demand is high, land is dear. If demand is low, land is cheap. In the entrepreneurial firm, the cost (price) of the entrepreneur varies in exactly the same way. If demand for the firm's special talents or products is high, the entrepreneur's payments are high. If demand is low, the entrepreneur's payments are low. Whereas centuries of thought have gone into rationalizing the curious behavior of land rent, when faced by the same curious behavior in an unpriced scarce

[25] We are hardly the first to point this out. See, for example, Thirlby (1946/1981) and Robbins (1934/1981).

[26] It does not mean it should be closed because that decision requires altering the opportunity costs to reflect the liquidation value of resources employed.

resource, Friedman and economists in general waver. We should not.

The resistance to saying Cost = Revenue arises from a sense of lost analytical power: what is the meaning of cost analysis if cost is revenue? Actually, nothing is lost by this identity. The elements economists and analysts call 'cost' are payments for various purchased inputs. Some of these payments are for inputs purchased by the ream, barrel, or hour, whereas others are for resources like trucks and buildings that are purchased periodically. The elements economists and analysts call 'profit' are payments for economically scarce unpriced inputs owned by the firm. What is gained is the throwing off of error.

Eva and the Expected Return Fallacy

False concepts of cost and opportunity cost are not harmless. For example, consider the popular EVA (economic value added) methods of business performance evaluation. This trademarked technique is marketed by Stern Stewart & Co. The measure is defined as EVA = NOPAT - [Capital Employed] $\times$ WACC, where NOPAT is operating profit after taxes for the unlevered firm, CapitalEmployed is obtained from book values, and WACC is the weighted average cost of capital appropriate to the firm or business. This number supposedly reveals whether or not the business is 'creating value' for the shareholder. The logic proposed by the method's advocates is that 'EVA is an estimate of true 'economic' profit, or the amount by which earnings exceed or fall short of the required minimum rate of return that shareholders and lenders could get by investing in other securities of comparable risk. [27]

The calculation of EVA actually performs a binary comparison between (a) receiving NOPAT as an annuity and (b) closing the firm and collecting the amount CapitalEmployed. If EVA < 0, you know that the option of closing the firm is better than receiving NOPAT each year. There is no indication that closing the firm is the best course of action. If EVA > 0, you know that there is no gain to closing the firm. A positive EVA does not mean the firm is creating shareholder value. It only means that closing the firm does not!

To honestly determine whether or not shareholders would be better off if the business is operated or closed, one must compare operating value to salvage value. The later is likely to be much smaller than its book value. To determine if a different strategy is appropriate, you must compare the existing value of the business to its expected value under the new strategy. Not easy? No one said strategy is easy.

If the EVA calculation were based on the market value of nonscarce resources, rather than the book value of all purchased resources, it might be an estimate of payments to scarce resources. But that is neither the method nor the claim. The claim is that EVA adjusts for the opportunity cost of capital and, as such, illustrates what we call the expected return fallacy. This line of reasoning is a cousin of the sunk cost fallacy. It holds that because a past investment has been made, it should earn a normal rate of return. Arguments leading to the expected return fallacy usually have

the form: 'If the investment does not earn a normal return, the investment capital will (or should) be withdrawn.' We saw this language in the Stiglitz quote in the section on Standard Textbook Doctrine; it also appears in a large number of books on strategy. Marshall first pointed out how rents to specialized resources lead to this error in reasoning. In his parable of the scarce, valuable, and indestructible meteoric stones, he noted:

> Since anyone, who bought stones, would take them from other producers, his purchase would not materially affect the general relations of demand for the

services of the stones to the supply of those services. It would not therefore affect the prices of the stones; which would still be the capitalized value of the services which they rendered in those uses, in which the need for them was the least urgent: and to say that the purchaser expected normal interest on the price which represented the capitalized value of the services, would be a circular statement that the value of the services rendered by stones is governed by the value of those very services (Marshall, 1920: 417, underline emphasis added)

To see the fallacy, consider parcel B of sunflower land. A farmer buys 1 acre of this land, anticipating simple rent payment of 10 per period. The interest rate is 6 percent, so the farmer pays $10/0.06 = 166.67$ for the acre. However, after some time, the demand for sunflowers unexpectedly falls because ρ fell from 20 to 18. The price of sunflowers drops, and, consequently, the rent on parcel B land falls from 10 to 8.9. The integrated farm now computes EVA (based on the book value of the land) as $8.9 - 166.67 * 0.06 = 8.9 - 10 = -1.1$. The consultants, saying 'value is being destroyed,' recommend sale. Such advice is foolish. [28] The original historical investment is not recoverable because the market value of the land has fallen right along with the drop in rent (or NOPAT). The original capital cannot be withdrawn; part of the investment is simply lost. Investment does not cease-farmers continue to buy land, albeit at a lower price. It is fallacious reasoning to claim as a 'cost' one's hoped for return on capital that is specialized and/or limited in supply.

From Costs to Payments

We have analyzed the nature of economic rents and the conceptual problems associated with the ideas of economic cost and opportunity cost. Our recommendation is that a payments-oriented perspective aids in avoiding conceptual confusion and logical fallacies. The Payments Perspective also focuses attention on the central issue in the resource-based view: the imputation of value to resources. The specific remedies we suggest are threefold: I. Define (expected) economic profit to be zero. Economic profits of zero are fully consistent with making a great deal of money. When economic profit is zero, net revenue is fully attributed (imputed) to resources. A positive economic

[27] http://www.sternstewart.com/evaabout/whatis.shtml.

[28] Were the consultants using market values instead of historical costs, a negative EVA might indicate inefficiency in operations. Of course, it might also simply mean that NOPAT is expected to increase in the future.

profit (in equilibrium) marks a lack of full imputation. Given the resource-based view's interest in a full accounting of resources as the source of wealth, it should define economic profit to be zero as an identity. II. Because economic profit is zero, Revenue $\equiv$ Cost. We prefer the word 'payment' to the word 'cost' because it carries less historical baggage. Payments are of three basic types: (1) payments for commodity inputs (PCI) in perfectly elastic supply; (2) payments for commodity resources (PCR) such as delivery trucks, file cabinets, office buildings, and so on; (3) payments for scarce resources (PSR). Rent sensitivity analysis determines a resource's category. Revenue $= \mathrm{PCI} + \mathrm{PCR} + \mathrm{PSR}$. Net revenue = PCR+ PSR, the payments to all resources. III. Payments for scarce resources can be divided into groups of payments for separable bundles. Two bundles of resources A and B are separable if and only if the value of their combined use is the sum of their separate use values: $\mathrm{V(A)} + \mathrm{V(B)} = \mathrm{V(A \cup B)}$. That is, separability implies a lack of synergy or co-specialization. Not all of these resources are necessarily owned by the firm. The division of payments within the bundle depends upon the pattern of cospecialization and bargaining power. The capitalized stream of payments for the services of a resource is its value; the total payment (cash flow) for all the resources in a firm is the value of the firm. The objective of the firm is to maximize its wealth by adjusting the ways in which it uses and combines resources and by its purchase and sale of resources.

Figure 6 illustrates the structure of the 'payments' logic. Note that within the payments approach competition induces the firm to maximize PSR. Competition drives the creation and configuration of resources that create economic surplus and drives the struggle over the division of the surplus. The welfare implication of this viewpoint is direct: entrepreneurial energy and innovation are starkly biased towards the creation of those surpluses which can be appropriated by the innovator.

Conclusions

We have argued that essential building blocks of the neoclassical theory of the firm are fragile. To step around the problems inherent in partial equilibrium theory's 'next-best use' doctrine, which is designed to justify the construction of supply curves, we have introduced rent sensitivity analysis and the payments perspective. Rent sensitivity analysis examines how simple rent payments vary with changes in demand. The payments perspective also puts simple rents in the forefront, recognizing that wealth consists of payments to scarce resources. We also emphasize that wealth exists whether or not the rent-generating resources are priced in factor markets.

Readers of this paper will see their way through the issues raised by Pindyck and Rubinfeld's explanation of the 'cost' of a good river location:

One firm is located on a river and can ship its products for $\$10,000$ a year less than the other firm, which is inland. Then the $\$10,000$ higher profit of the first firm is due to the $\$10,000$ per year economic rent associated with its river location. The rent is created because the land along the

river is valuable, and other firms would be willing to pay for it. Eventually, the competition for this specialized factor of production will increase its value to $\$10,000$. Land rent-the difference between $\$10,000$ and the zero cost of obtaining the land-is also $\$10,000$. Note that while the economic rent has increased, the economic profit of the firm on the river has become zero. (Pindyck and Rubinfeld, 1992: 268)

Pindyck and Rubinfeld exercise the language of neoclassical theory. Unfortunately, this exercise hides the essential strategic forces at work. Using the payments perspective, economic profits are always zero, so we ignore that aspect of the argument. Because all wealth resides in factor values, we focus on this issue rather than the spurious issue of economic profit or the intellectually exhausting question of the 'cost' of the land. The essential point is that an innovation (or new use) has increased the value of the land by $\$10,000$ per year. Who appropriates this increase in rent? If the innovation is specific to the location, the value of other land is unaffected. If the innovating business person were clever enough to buy the land before innovating, she captures the capital gain in land values. If the innovation is specific to this entrepreneur (nonimitable), then the fundamental transformation turns the situation into a bilateral bargaining problem between the innovator and the landowner. If the innovation is specific to the location, but fully imitable by other entrepreneurs and firms, the landlord will increase the rent as soon as the innovation is established. The landlord, who could be the innovator, captures the value.

Research on the resource-based view is on the right track. Let us not permit errors in the microfoundations of neoclassical theory to slow down our progress.

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References

Alchian A. 1987. Rent. In The New Palgrave: A Dictionary of Economic Theory and Doctrine, Vol. 4, Eatwell J, Milgate M, Newman P (eds). Macmillan: London; 141-143.

Barney JB. 1986. Strategic factor markets: expectations, luck and business strategy. Management Science 32: 1231-1241.

Barney JB. 2001. Resource-based theories of competitive advantage: a ten-year retrospective on the resourcebased view. Journal of Management 27: 643-650.

Becker GS. 1971. Economic Theory. Alfred A. Knopf: New York.

Besanko D, Dranove D, Shanley M. 2000. Economics of Strategy, 2nd edn. Wiley: New York.

Bowman E. 1974. Epistemology, corporate strategy, and academe. Sloan Management Review 15: 35-50.

Buchanan JM. 1969. Cost and Choice. University of Chicago Press: Chicago, IL.

Buchanan JM. 1987. Opportunity cost. In The New Palgrave: A Dictionary of Economic Theory and Doctrine, Vol. 3, Eatwell J, Milgate M, Newman P (eds). Macmillan: London; 718-721.

Caves RE. 1980. Industrial organization, corporate strategy and structure. Journal of Economic Literature 58: 64-92.

Coase RH. 1938. Business organization and the accountant. The Accountant (October-December). Reprinted in Solomons D (ed). 1952. Studies in Costing. Sweet & Maxwell: London; 105-158.

Cohen KJ, Cyert RM. 1965. Theory of the Firm: Resource Allocation in a Market Economy. Prentice-Hall: Englewood Cliffs, NJ.

Dierickx I, Cool K. 1989. Asset stock accumulation and sustainability of competitive advantage. Management Science 35: 1504-1511.

Friedman M. 1976. Price Theory. Aldine: New York.

Hirshleifer J. 1980. Price Theory and Applications, 2nd edn. Prentice-Hall: Englewood Cliffs, NJ.

Kaldor N. 1934. The equilibrium of the firm. Economic Journal 44: 60-76.

Lippman SA, Rumelt RP. 2003. A bargaining perspective on resource advantage. Strategic Management Journal 24(11): (forthcoming) DOI: 10.1002/smj.345.

Mahoney JT. 2001. A resource-based theory of sustainable rents. Journal of Management 27: 651-660.

Mahoney JT, Pandian JR. 1992. The resource-based view within the conversation of strategic management. Strategic Management Journal 13(5): 363-380.

Marshall A. 1920, reprinted 1946. Principles of Economics, 8th edn. Macmillan: London.

Milgrom P, Roberts J. 1992. Economics, Organization and Management . Prentice-Hall: Upper Saddle River, NJ.

Mill JS. 1871, reprinted 1987. Principles of Political Economy. Augustus M. Kelley: Fairfield, NJ.

Montgomery CA, Wernerfelt B. 1988. Diversification, Ricardian rents, and Tobin's q. Rand Journal of Economics 19: 623-632.

Nelson RR. 1957. Increased rents from increased costs: a paradox of value theory. Journal of Political Economy 65: 387-393.

Pareto V. 1897. Cours d'Economie Politique, 2 vols. Libraire de l'Universitie: Lausanne.

Peteraf MA. 1993. The cornerstones of competitive advantage: a resource-based view. Strategic Management Journal 14(3): 179-191.

Pindyck RS, Rubinfeld DL. 1992. Microeconomics, 2nd edn. Macmillan: New York.

Ricardo D. 1821/1951. On the principles of political economy and taxation. In The Works and Correspondence of David Ricardo, 1821, 3rd edn, Sraffa P (ed). Cambridge University Press: Cambridge, U.K.

Robbins L. 1932. Essay on the Nature and Significance of Economic Science. Macmillan: London.

Robbins L. 1934/1981. Remarks upon certain aspects of the theory of costs. In LSE Essays on Cost, Buchanan JM, Thirlby GF, (eds). New York University Press: New York; 19-42.

Robinson J. 1933/1969. The Economics of Imperfect Competition, 2nd edn. Macmillan: London.

Rumelt RP. 1984. Toward a strategic theory of the firm. In Competitive Strategic Management, Lamb R (ed). Prentice-Hall: Englewood Cliffs, NJ; 556-570.

Rumelt RP. 1987. Theory, strategy, and entrepreneurship. In The Competitive Challenge, Teece DJ (ed). Ballinger: Cambridge, MA; 137-158.

Samuelson PA. 1947/1974. Foundations of Economic Analysis. Atheneum: New York.

Samuelson PA. 1959. A modern treatment of the Ricardian economy: I. The pricing of goods and of labor and land services. Quarterly Journal of Economics 73: 1-35.

Samuelson PA, Nordhaus WD. 2001. Economics, 17th edn. McGraw-Hill/Irwin: New York.

Shepherd A. 1970. Economic rent and the industry supply curve. Southern Economic Journal 37: 209-211.

Shubik M. 1971. Pecuniary externalities: a game theoretic analysis. American Economic Review 61: 713-718.

Smith A. 1776/1976. An Inquiry into the Nature and Causes of the Wealth of Nations, Cannan E (ed). University of Chicago Press: Chicago, IL.

Stigler GJ. 1966. The Theory of Price. Macmillan: New York.

Stiglitz JE. 1997. Principles of Microeconomics, 2nd edn. W. W. Norton: New York.

Teece DJ. 1986. Profiting from technological innovation: implications for integration, collaboration, licensing and public-policy. Research Policy 15: 285-305.

Thirlby GF. 1946/1981. Economists' cost rules and equilibrium theory. In LSE Essays on Cost, Buchanan JM, Thirlby GF (eds). New York University Press: New York; 273-288.

Varian H. 2003. Intermediate Microeconomics: A Modern Approach (6th edn). W. W. Norton: New York.

Wernerfelt B. 1984. A resource-based view of the firm. Strategic Management Journal 5(2): 171-180.

Wernerfelt B. 1995. The resource-based view of the firm: 10 years after. Strategic Management Journal 16(3): 171-174.

Wessel RH. 1967. A note on economic rent. American Economic Review 57: 1221-1226.

Wicksteed PH. 1910. The Common Sense of Political Economy, Including a Study of the Human Basis of Economic Law. Macmillan: London.

Wieser F von. 1893. Natural Value, Smart W (ed). Macmillan: London.

Williamson O. 1985. The Economic Institutions of Capitalism. Free Press: New York.

Winter SG. 1987. Knowledge and competence as strategic assets. In The Competitive Challenge, Teece DJ (ed). Ballinger: MA.

Winter SG. 1995. Four Rs of profitability: rents, resources, routines, and replication. In Resource-Based and Evolutionary Theories of the Firm, Montgomery C (ed). Kluwer: Boston, MA; 147-178.

Worchester DA. 1946. A reconsideration of the theory of rent. American Economic Review 36: 258-277.

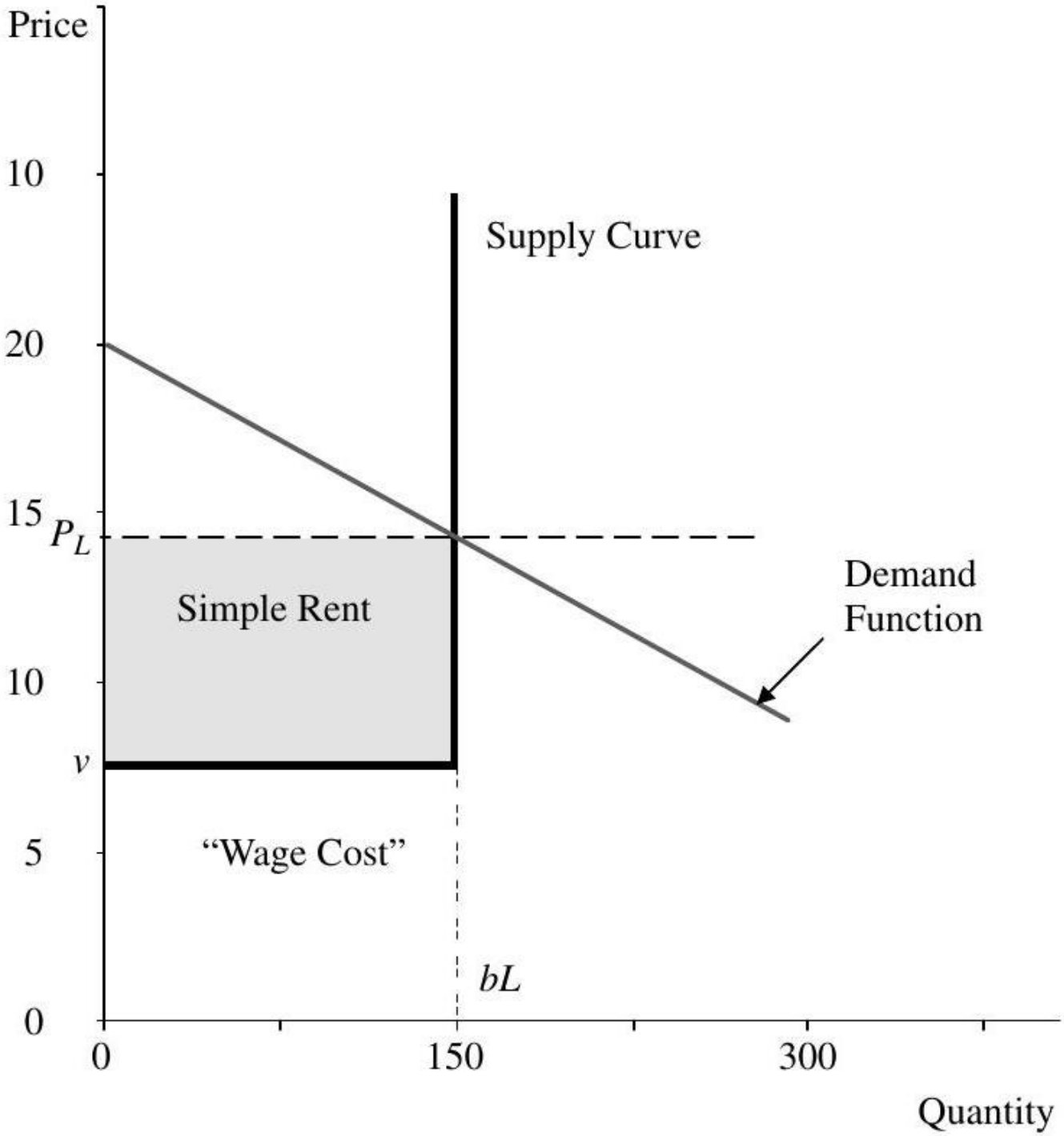
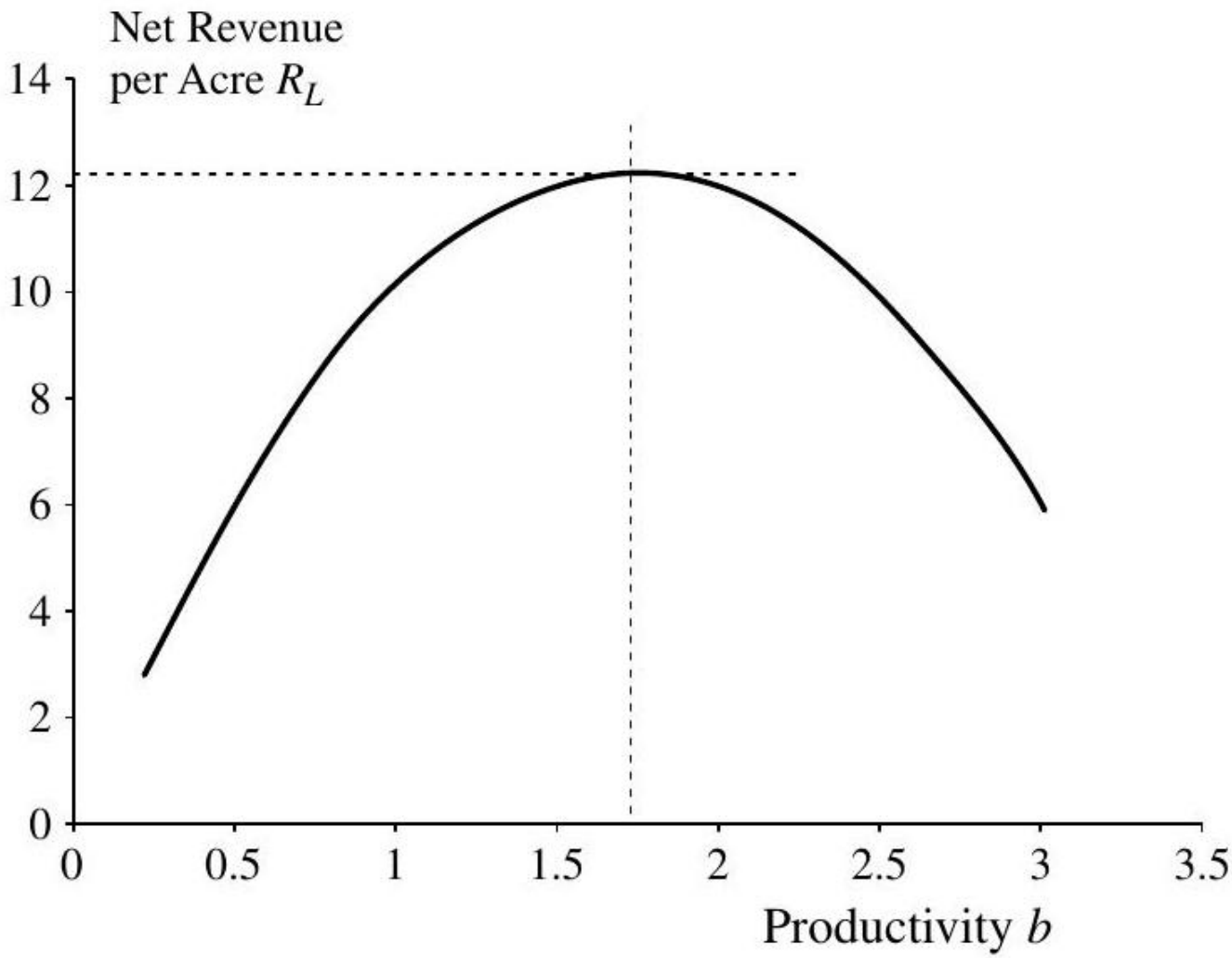


Figure 1. Supply curve for sunflower land

Theorem 3: If $b < b^*$ there are no gains to monopolization. If $b > b^*$, then a monopoly owner of the land will set production at Q^* and



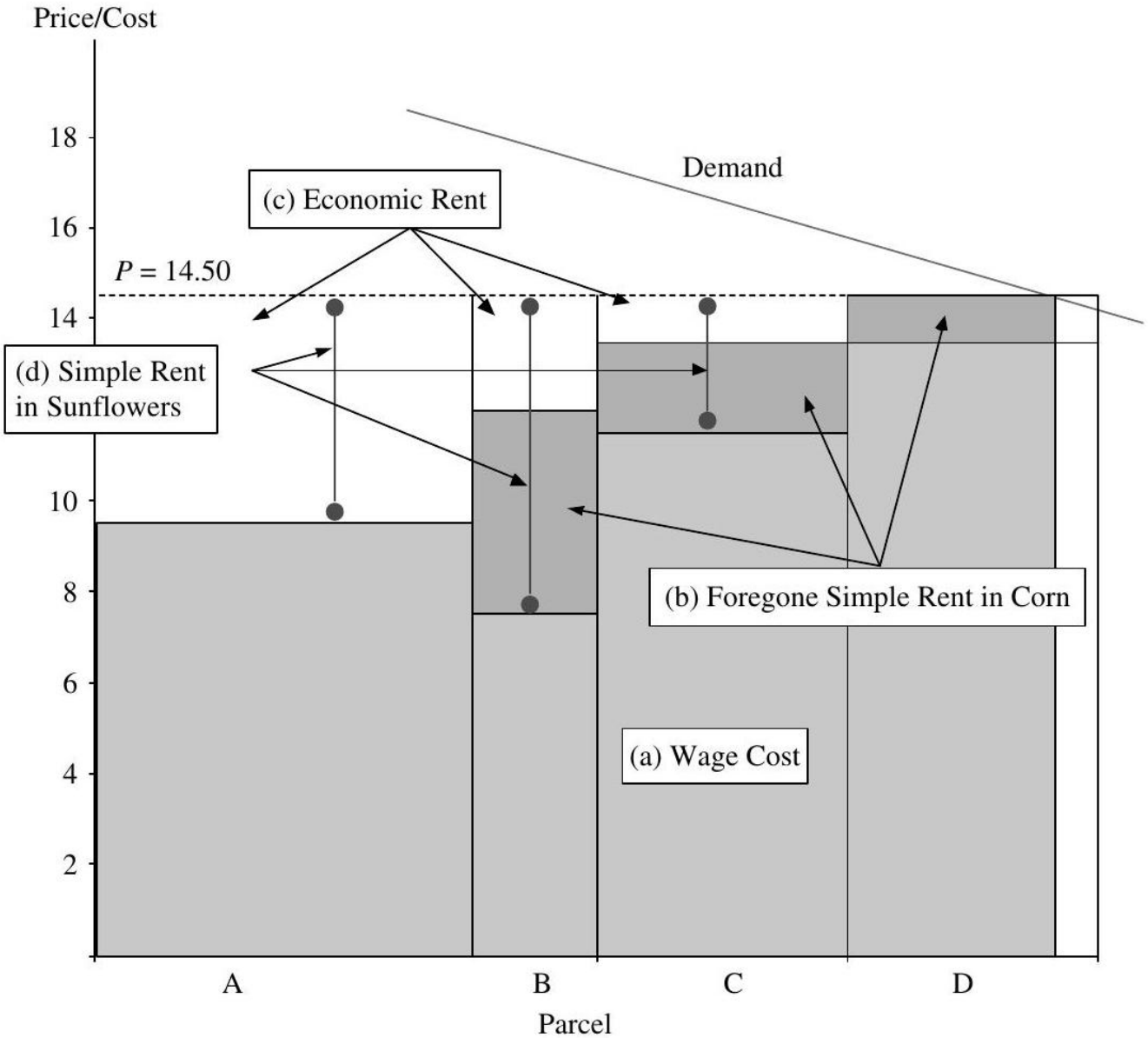


Figure 3. Sunflower industry supply curve with four parcels

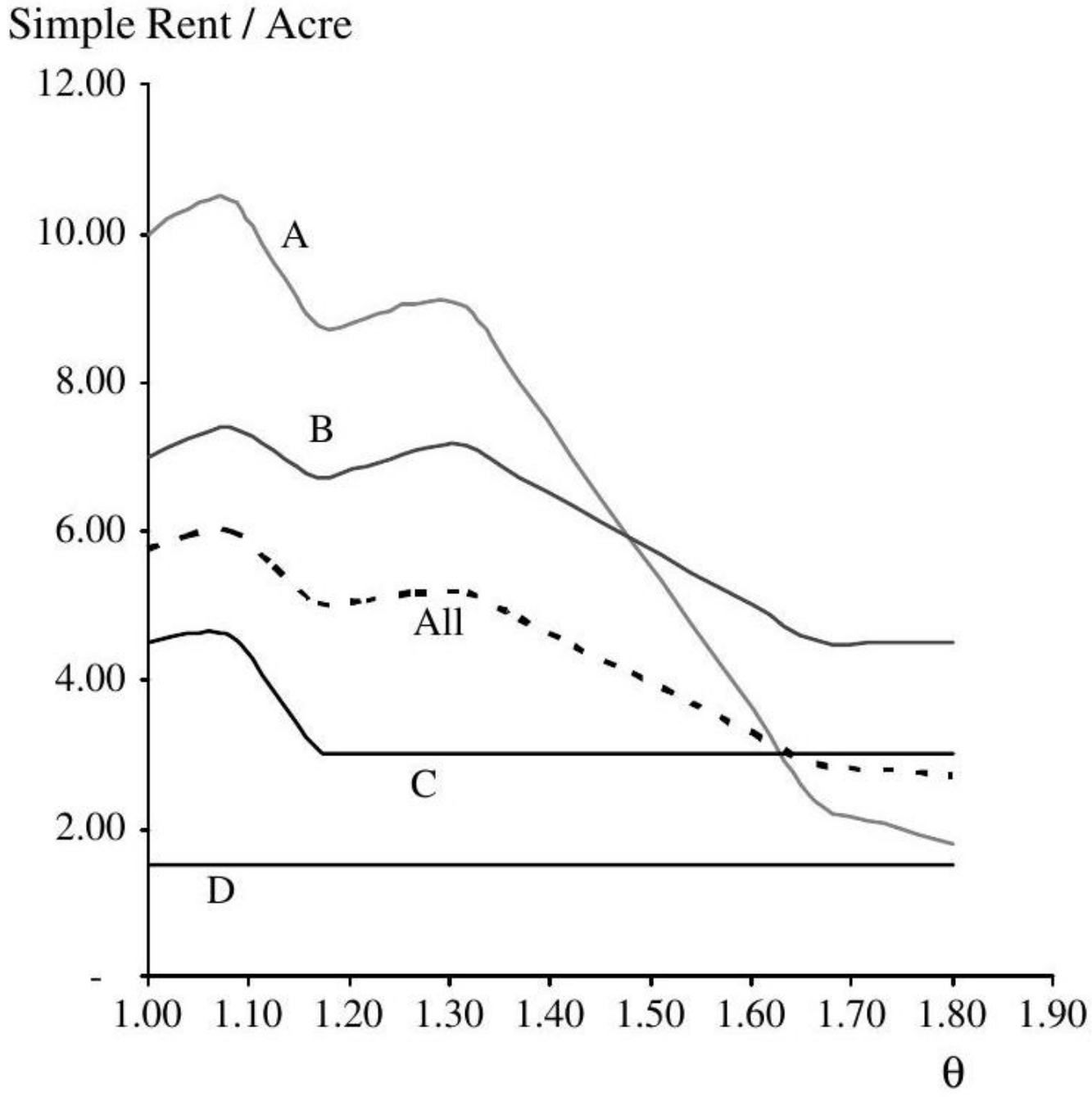


Figure 4. Simple rents on parcels A-D vs. wage cost factor θ

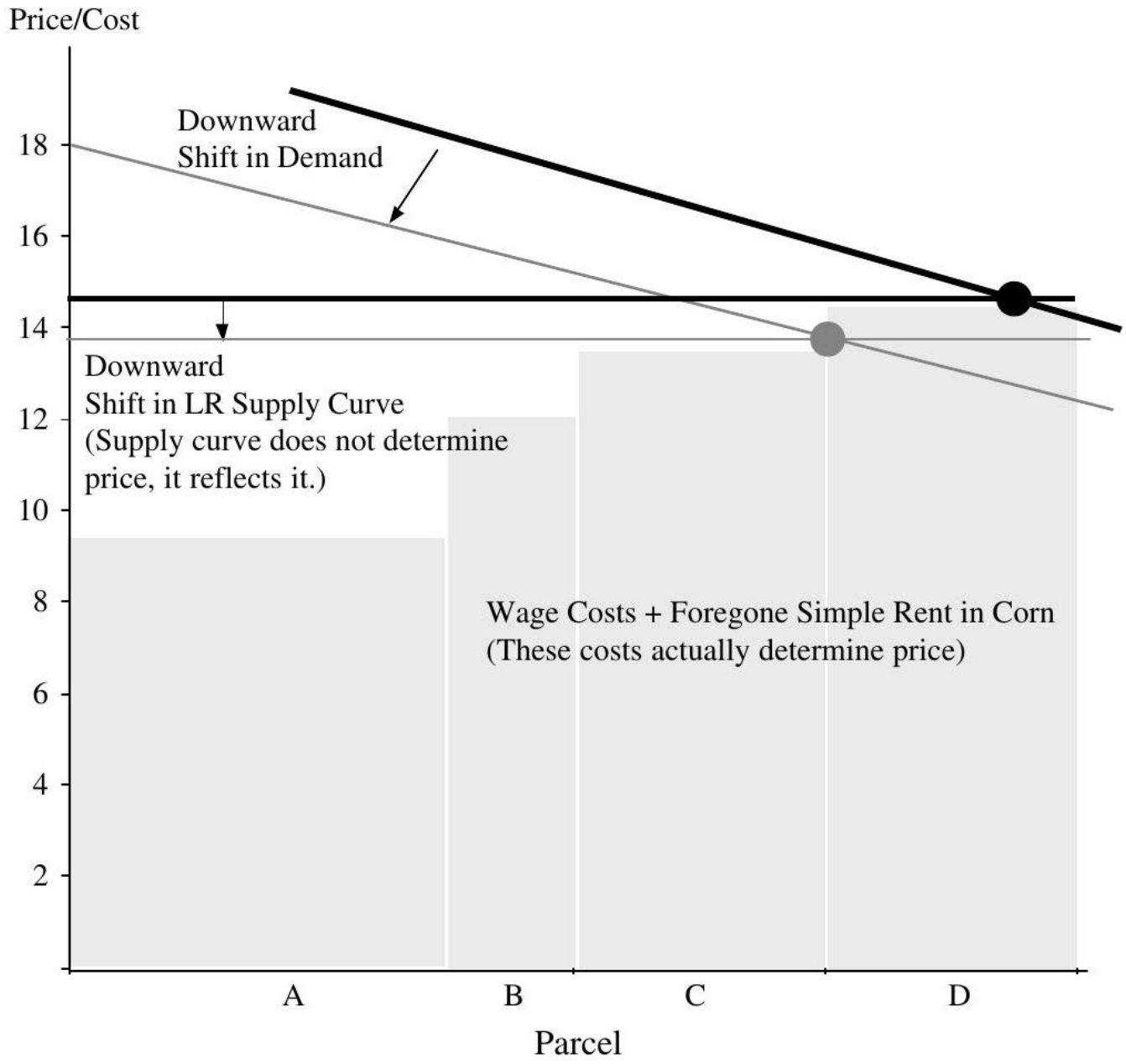


Figure 5. Long-run sunflower demand and supply

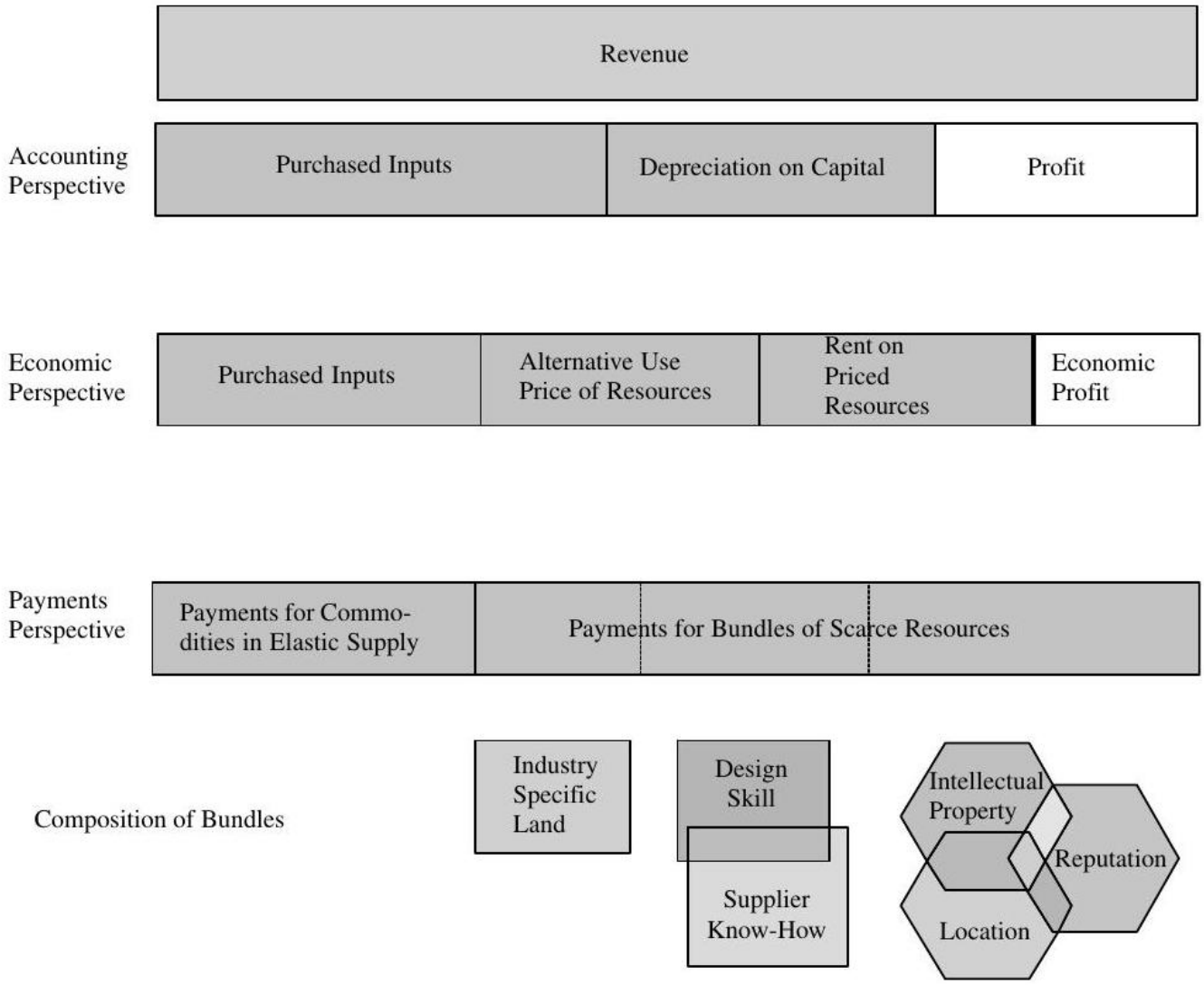


Figure 6. Payments Perspective